

Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.keaipublishing.com/jtte

Review Article

Public transport inclusion and active aging: A systematic review on elderly mobility

Na Zhang ^{a,b}, Qi Yang ^{c,*}^a School of Humanities, Chang'an University, Xi'an 710064, China^b School of Transportation Engineering, Chang'an University, Xi'an 710064, China^c School of Economics and Management, Chang'an University, Xi'an 710064, China

HIGHLIGHTS

- A multidisciplinary review on the theoretical underpinnings of transport inclusion is presented.
- The mobility dilemma and public transport needs of older adults are summarized.
- The mechanisms that transport influences active aging are analyzed.
- A research framework for transport including actively aging societies is proposed.
- Potential challenges and recommendations for future transport inclusion research are presented.

ARTICLE INFO

Article history:

Received 22 February 2024

Received in revised form

25 March 2024

Accepted 9 April 2024

Available online 10 April 2024

Keywords:

Transportation engineering

Elderly mobility

Transport-related social exclusion

Transport inclusion

Active aging

ABSTRACT

Global aging has brought attention to addressing the elderly mobility dilemma, and a commitment to address inclusive transport in aging. Transport inclusion for older adults aims at establishing an age-friendly transport system. As a direct policy goal and promoting active aging as an indirect social goal, to help older populations gain access to transport and social opportunities. Based on the analysis of scientific knowledge mapping over the past 30 years, this study visualizes the hotspots and trends in elderly mobility research. Through a qualitative analysis of 138 papers, this paper reviews the theoretical foundations for transport inclusion and three themes of elderly mobility research, proposes a framework for public transport (PT) inclusion research as well as future research directions. The study suggests that transport inclusion can help enhance elderly health, social participation, and subjective well-being. It can improve their travel opportunities for active aging in four ways: opportunity accessibility, physical accessibility, age-friendly informationalization, and cost affordability. Future research should pay attention to examining the mechanism of perceived transport inclusion on active aging, expanding the multilevel research space of the capability approach (CA), and exploring the subjective-objective integrated research model for PT inclusion. This research contributes to the overall knowledge of the theoretical and empirical researches on PT inclusion for the elderly and promotes the integrated application of interdisciplinary knowledge in the field.

© 2024 Periodical Offices of Chang'an University. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

* Corresponding author.

E-mail addresses: zhangna@chd.edu.cn (N. Zhang), yangqi@chd.edu.cn (Q. Yang).

Peer review under responsibility of Periodical Offices of Chang'an University.

<https://doi.org/10.1016/j.jtte.2024.04.001>2095-7564/© 2024 Periodical Offices of Chang'an University. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

Population aging is a global problem worldwide. It is projected that by 2050, 23% of the world's population will be aged 60 years or older (United Nations, 2017). The WHO proposes that active aging aims to optimize the health, participation and security of older people to improve their life quality. Since then, active aging has been recognized worldwide as a valuable policy goal for addressing globalized aging. Since the release of the active aging index (AAI) by the European Commission and the United Nations Economic Commission for Europe (UNECE) in 2012, there has been a sustained upward trend in the focus on the impact of elderly mobility (Pantelaki et al., 2021). Among the active aging indices proposed by different scholars, transport is considered as an important indicator for achieving active aging (Li et al., 2020a, 2020b). In particular, public transport (PT) inclusion is crucial for active aging. Because of age and physical factors, many older adults stop driving and become highly dependent on PT (Whelan et al., 2006). PT is an active travel mode affecting the physical and mental health of older adults, enabling them to maintain social networks and thus reduce social isolation (Rambaldini-Gooding et al., 2021). However, the elderly is experiencing mobility dilemmas in four aspects such as PT accessibility, facility availability, information comprehensibility, and cost affordability. In societies with aging demographics, transport policy initiatives will become unsustainable if insufficient attention is paid to the transport dilemma for older people (Yigitcanlar et al., 2015). Despite the positive effects of PT inclusion on elderly life quality and aging society development, older people's PT needs have not yet received sufficient attention (Adorno et al., 2018). Transport inequality is worse for the elderly than for the total population on average, and compared to other travel disadvantaged groups (e.g., low-income, households without a car, other age groups), the elderly have the highest level of inequality in PT resource distribution (Ermagun and Tilahun 2020; Ricciardi et al., 2015).

The elderly are a highly heterogeneous disadvantaged traveler group. Research on PT inclusion for them requires a synthesis of relevant knowledge from health sciences, transport sciences, and social sciences. In the field of health sciences, active aging has been discussed in depth as an important academic concept in the study on the elderly, and the relationship between transport and active aging has been widely mentioned. However, the research value for this concept in the transport field has not yet gained attention, and there is a lack of transport policies for older people (Patil et al., 2022). It is necessary to discuss PT inclusion for them in the context of the goal of active aging. Four important themes have been developed in the field of elderly mobility research: elderly travel research, transport and aging relationship research, transport exclusion research, and inclusive transport research. A systematic review on research results in the elderly mobility field, from the perspective of PT

inclusion, can strengthen the integration and application for existing results in the PT inclusion research field.

Existing research findings have revealed elderly mobility dilemma and further discussed the exclusive or inclusive results of transport resource distribution in the social sphere. These studies have led to reflections on the limitations in the efficiency prioritization principle for traditional transport policy assessment, and the important value of transport inclusion for disadvantaged groups has been recognized (Stanley et al., 2011a). Existing research findings have identified four dimensions in transport exclusion and have provided a theoretical basis for the design of a policy framework for elderly transport inclusion. However, there is a lack of systematic research on transport inclusion in the existing literature. No studies on the topic of inclusive transport have explicitly discussed the framework, objectives and elements of a transport inclusion system. Existing studies have produced a large number of empirical findings in terms of accessibility, physical accessibility, age-friendly informationalization and cost affordability, but no study has systematically discussed the value of these findings in building a policy system for elderly PT inclusion.

The purpose of this paper are, through a structured review of papers on elderly mobility research topics and a qualitative interpretation of research evidences, to integrate theoretical and empirical research findings related to PT inclusion with the goal of promoting active aging as a transport policy, to propose a policy framework for elderly PT inclusion, and to explore the future research space. This paper contributes to a systematic understanding across interdisciplinary knowledge outcomes for vulnerable elderly travelers, critically analyzes important innovative findings, bottlenecks, and research directions for transport inclusion, and provides insights for theoretical research and policy design on transport inclusion.

The structure of this paper consists of four sections. Section 2 uses an econometric and visual analysis on 30 years literature data to reveal the research lineage and hot trends in the elderly mobility field. Section 3, from the multidisciplinary perspective, examines the theoretical foundations of transport inclusion and the mechanisms by which transport generates social exclusion or social inclusion. Through a critical analysis on 103 selected literatures, Section 4 summarizes three research themes on elderly PT mobility: characteristics of elderly PT travels, PT's impacts on active aging, and transport exclusion factors affecting elderly mobility. Based on Sections 4, 5 proposes a research framework for elderly PT inclusion, which are opportunity accessibility, physical accessibility, age-friendly informationalization, and cost affordability, and summarizes the research results in these areas. Section 6 proposes a future research space for transport inclusion. In view of the fact that the representative terms have similar meanings but lack consistent names in different papers, this paper lists the detailed descriptions of the representative terms (Table 1).

Table 1 – Detailed descriptions of the representative terms.

Term	Explanation
Negative aging	Under the influence of disengagement theory, negative aging is viewed as a decline in biological attributes and a gradual disengagement from social roles, which is universal and inevitable (Foster and Walker, 2014).
Active aging	Active aging is a policy goal proposed by the WHO to address global aging. It aims to optimize the health, participation and security for older people, enabling them to realize their potential for physical, social and spiritual well-being and to participate in society in accordance with their needs, aspirations and abilities, while providing them with adequate protection, security and care when they need assistance (World Health Organization, 2002).
Social inclusion	The Europe Commission defines social inclusion as a tendency to enable people at risk of poverty or social exclusion to have the opportunity to participate fully in social life, and thus enjoy an adequate standard of living considered normal in the society in which they live (Europe Commission, 2020). The Australian Social Inclusion Board defines social inclusion as people having the resources, opportunities, and capabilities to participate in the normal activities of a society (Australian Social Inclusion Board, 2012). Social inclusion is especially concerned about people or groups of people who are at risk of deprivation, segregation or marginalization, dedicated to improving the conditions for their participation in society (Paraschiv et al., 2021).
Social exclusion	Social exclusion is the unequal access to some development opportunities and rights, for excluded subjects, as a result of unequal power relations or unequal distribution of resources, with negative long-term consequences for the well-being of both individuals and society (Popay et al., 2008). Social exclusion highlights the importance of mobility in modern cities as a key factor in people's lives, noting that insufficient mobility can cause less accessibility to opportunities, services or social networks (Kenyon et al., 2002).

Table 1 – (continued)

Term	Explanation
Transport-related social exclusion (TRSE)	TRSE is the process by which, at a normal standard of living, disadvantaged groups are marginalized or segregated from society by inhibiting their ability to participate in daily activities through transport, resulting in a lack of opportunity to participate in activities generally agreed upon by society (Mackett, 2014b). TRSE can occur in groups through socio-demographic characteristics such as persons with disabilities, older adults, low-income people, women, children, aboriginal people, and so on.
Transport exclusion	Transport exclusion defines those transport factors that bring social exclusion. It is a multidimensional concept describing the elderly mobility dilemma, including opportunity accessibility exclusion, physical accessibility exclusion, information exclusion, economy exclusion.
Transport inclusion	Transport inclusion is the implementation for inclusive transport policies, with transport equity as the fundamental resource distribution concept, so that disadvantaged travelers can overcome personal and resource constraints in accessing transport opportunities to ensure that they have the basic capabilities necessary for their survival and personal development.

2. Scientific knowledge mapping analysis on the literature

2.1. Data sources

Mapping knowledge domain (MKD), known as knowledge domain visualization or knowledge domain mapping in the library and intelligence community, is a series of different graphs showing the process of knowledge development and structural relationships, describing the knowledge resources and their carriers with visualization techniques, mining, analyzing, constructing, mapping and displaying the knowledge and the interconnections among them. In this study, the SCI and SSCI databases in the Core Collection Web of Science were used as the source of literature data, and the time-frame was from 1990 to December 2022 to retrieve academic results in the field of elderly mobility research. The first search keywords were older/elderly mobility, older/elderly travel, active aging, inclusion/inclusive transport, exclusion/exclusive transport, social inclusion/exclusion, barrier free transport. After focusing on the disciplines of transport, environmental sciences, sociology, gerontology, and regional urban planning,

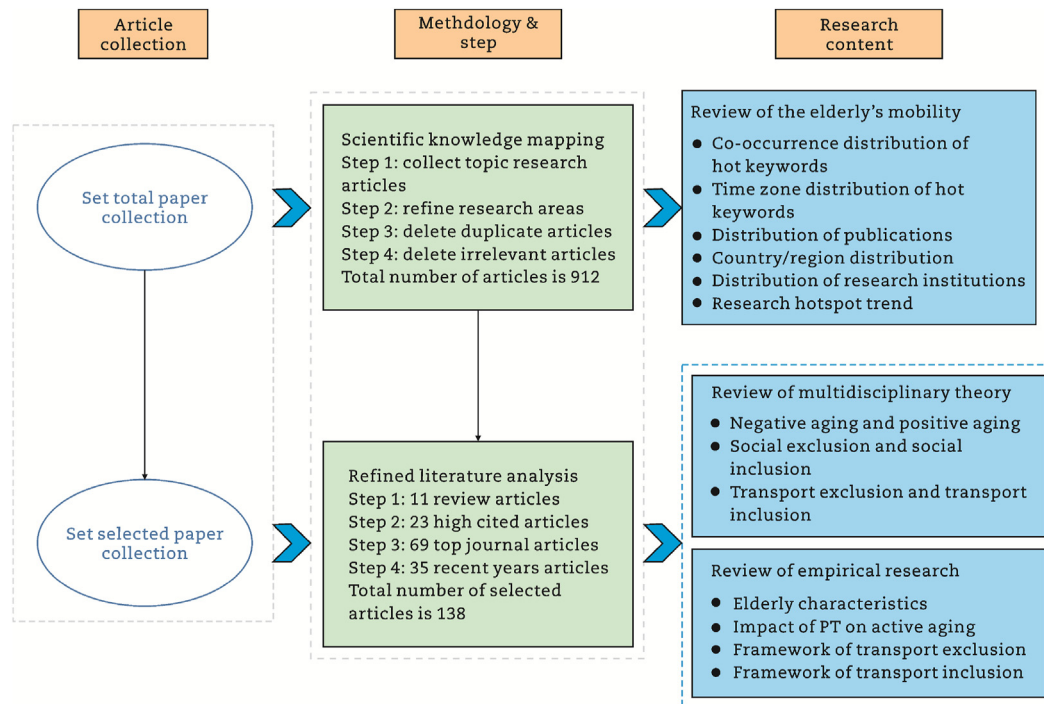


Fig. 1 – Flowchart of the systematic review process.

1020 English papers on elderly mobility were obtained, and after removing duplicates and reading the abstracts to exclude irrelevant papers, 912 papers were obtained to form the primary dataset.

The primary literature dataset was analyzed by scientific knowledge mapping using CiteSpace to comprehensively review the research hot spots and trends in the field of elderly mobility research over the past 30 years. The study included: co-occurrence distribution of hot keywords, time zone of hot keywords, distribution of publications, distribution of countries/regions, distribution of research organizations, and hot research trends.

Based on the reading of the full texts of the primary literature, 138 papers were selected to constitute the selected dataset. The papers and their cited research results were analyzed, and the important themes of the theoretical and empirical research on mobility of the elderly were synthesized. The papers in the selected dataset were mainly from review papers, highly cited papers, Q1 journal papers with high publication volume, and papers published in the last five years (supplemental searches were conducted for 2022–2024). Fig. 1 provides details of the research steps.

2.2. Co-occurrence distribution of hot keywords

The literature in the primary dataset contained a total of 551 keywords. Keywords with more than 15 occurrences were regarded as hot keywords, and the total number of keyword occurrences was 1518. The keyword co-occurrence was mapped, revealing 234 nodes and 617 lines. Based on the

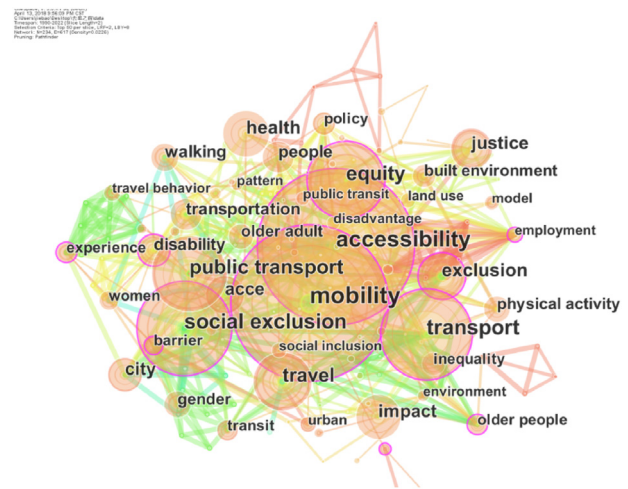


Fig. 2 – Distribution of keywords co-occurrence.

above hot keywords, the co-occurrence matrix was constructed and the co-occurrence knowledge of hot keywords was mapped (Fig. 2).

Through the analysis of high-frequency words (Table 2), it was found that the research hot spot at the level of transport theory was equity; the research hot spots at the level of transport practice were mobility, accessibility and public transport. From the time sequence of word frequency, social exclusion appeared in 2003, and social inclusion appeared in 2016.

Table 2 – High-frequency keyword statistics (word frequency ≥ 16).

Keyword	Centrality	Frequency	Year
Mobility	0.18	165	2008
Accessibility	0.12	164	2009
Social exclusion	0.19	101	2003
Transport	0.11	96	2007
Equity	0.13	83	2012
Public transport	0.08	80	2009
Travel	0.02	67	2003
Health	0.03	54	2012
Exclusion	0.12	53	2008
Justice	0.07	49	2012
Impact	0.05	46	2012
Acce.	0.09	45	2010
City	0.09	43	2003
Disability	0.13	39	2010
Transport	0.04	38	2010
Walking	0.03	36	2001
Gender	0.02	32	2009
Physical activity	0.03	30	2015
Built environment	0.04	28	2012
Older adult	0.06	28	2015
Policy	0.06	27	2010
Inequality	0.01	26	2018
Older people	0.11	26	2007
Transit	0.02	26	2008
Women	0.02	25	2001
Barrier	0.11	25	2008
Urban	0.00	19	2018
Employment	0.13	18	2012
Social inclusion	0.00	17	2016
Space	0.09	16	2018
Public transit	0.01	16	2018

2.3. Distribution of publications

The literature in the primary dataset involved 158 internationally recognized academic journals and 36 international academic conferences. The top 10 academic journals which the literature belonged to are detailed in [Table 3](#). Among them, Q1 journals account for 50% and Q2 journals account for 40%, indicating that the topic of mobility of the elderly is a favored research topic in high-level journals in the world.

Table 4 – Centrality value of region.

Country	Centrality	Frequency	Year
USA	0.08	103	2005
England	0.57	101	2003
Australia	0.42	60	2001
Canada	0.18	60	2010
China	0.03	46	2010
Netherlands	0.16	24	2011
Italy	0.01	20	2014
Germany	0.11	19	2008
Scotland	0.00	17	2001
Spain	0.02	16	2017
Chile	0.07	16	2013
Sweden	0.01	14	2012
New Zealand	0.05	13	2007
Belgium	0.02	13	2014
Brazil	0.08	12	2018
Norway	0.00	12	2009
Ireland	0.06	10	2014
India	0.04	10	2011

2.4. Country/region distribution

The primary dataset consists of papers from 38 countries and regions, 78% of which have contributed more than 5 papers and have been cited more than 10 times. [Table 4](#) shows the number of papers and citation status of these countries/regions. In [Fig. 3](#), the node size represents the number of papers published in the country/region, the node color represents the research direction of the literature, and the thickness of the line between the nodes represents the citation frequency of the two countries/regions. The countries with active research are USA (Frequency = 103), England (Frequency = 101) and Australia (Frequency = 60). Of these, England (Centrality = 0.57) has the most influential research results, followed by Australia (Centrality = 0.42), both of which have been gradually focusing on this area of research since 2001. The earliest three countries entered this research field are Australia (2001), Scotland (2001), and England (2003). [Table 5](#) lists the chronological order in which countries/regions entered the topic.

Table 3 – Top 10 academic journal.

Ranking	Journal	Number of papers	Citations	Citation proportion (%)	JCR category quartile	Journal citation indicator
1	Transportation Research Part A: Policy and Practice	47	921	97	Q1	5.594
2	Transport	21	676	95	Q1	5.192
3	Journal of Transport Geography	46	510	95	Q1	4.986
4	Transport Policy	28	253	82	Q1	4.674
5	Transportation Research Record	33	149	78	Q4	1.560
6	Sustainability	36	148	80	Q2	3.251
7	BMC Public Health	3	134	100	Q2	3.295
8	Journal of Transport & Health	28	125	67	Q2	2.796
9	Transportation Research Part D: Transport and Environment	17	114	76	Q1	5.495
10	International Journal of Sustainable Transportation	4	85	100	Q2	3.929

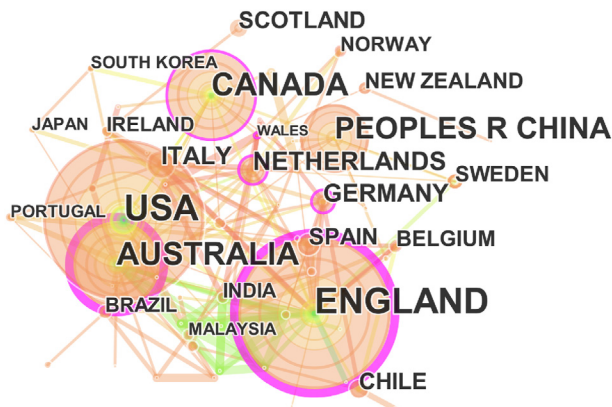


Fig. 3 – Region distribution.

Table 5 – Chronological order of the country/region entering the research field.

Year	Country
2001–2010	Australia, Scotland, England, USA, New Zealand, Germany, Norway, Canada, China
2011–2018	Netherlands, India, Sweden, Chile, Italy, Belgium, Ireland, Spain, Brazil

2.5. Distribution of affiliated research institutions

The literature in the primary dataset involved a total of 368 scholars from 260 research institutions, and there were 17 research institutions with more than 6 citations (Table 6). The influential research institutions (Centrality = 0.05) were University of Leeds, London's Global University, The University of Melbourne, University of Oxford, and the

Table 6 – Research institutions of cited references (Frequency ≥ 5).

Institution	Frequency	Centrality	Year
London's Global University	26	0.05	2011
University of Toronto	18	0.03	2010
University of Leeds	16	0.07	2013
The University of Melbourne	13	0.05	2017
Peking University	12	0.01	2018
University of Sydney	12	0.04	2011
Pontificia Universidad Catolica de Chile	10	0.00	2015
University of Oxford	9	0.05	2012
University of Auckland	9	0.01	2018
The University of Hong Kong	9	0.01	2018
McMaster University	9	0.01	2010
McGill University	9	0.04	2017
Monash University	7	0.00	2010
RMIT University	7	0.03	2018
University of Westminster	6	0.01	2018
Portland State University	6	0.02	2017
The Chinese University of Hong Kong	6	0.00	2018

academic influence of these institutions was still radiating outward in the last decade.

2.6. Research hot spot trends

The hot keywords time zone map can show the evolution of knowledge in the time dimension. The more literature there is in a certain time zone, more results, it means, are published in this time zone and so the field is in a prosperous period. Through the connecting line relationship between time zones, the inheritance relationship between time zones can be observed.

According to the time zone map of hot spot keywords (Fig. 4), it can be noticed that transport-related social exclusion (TRSE) research began to enter the research horizon from 2000, TRSE theme was formed in 2004, and transport-related social inclusion (TRSI) theme appeared in 2016. According to the analysis on mediated centrality, accessibility and mobility became the core theme areas in 2008–2010, and these two themes co-occurred more frequently as the basic research themes; equity is the core topic at the theoretical level, and the related thematic content receives a higher degree of attention, and the color of the circle shows that the theme is still a research hot spot in the current time. According to the theme connecting lines and colors, the connecting lines between the circles during 2008–2014 are relatively close, indicating that the medium-term theme has a close relationship with the early theme, and the recent theme has a close relationship with the medium-term theme.

The strongest citation bursts can be used to track the distribution of key nodes in different periods of time more intuitively and clearly. Fig. 5 shows the sudden increase or decrease of the citation frequency of the literature in a certain period of time, and these bursts directly reflect the significant shift of the research hot spots in a certain period

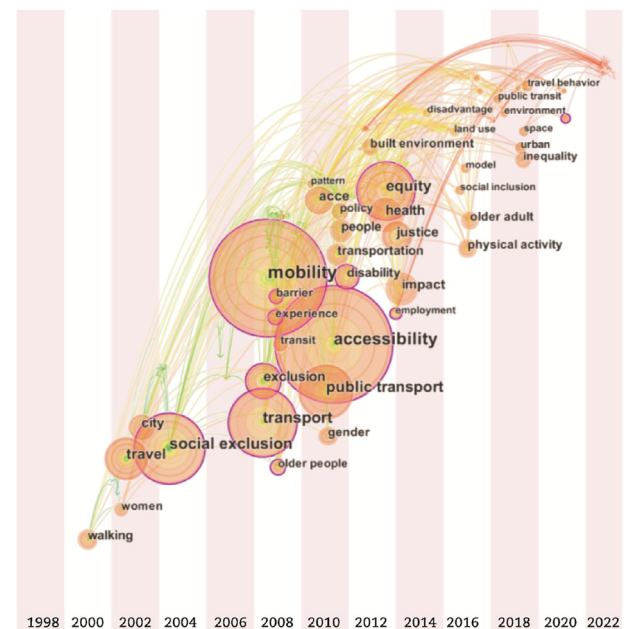


Fig. 4 – Time zone map of hot spot keywords.

Top 6 Keywords with the Strongest Citation Bursts



Fig. 5 – Keywords with the strongest citation bursts.

of time. Through the hot keywords emergence detection, this study found the emergence of six keywords, i.e., land use, disadvantage, employment, quality, older adult, and impact, in this research area.

3. Review of multidisciplinary theoretical basis

Elderly mobility research involves multiple disciplinary fields such as transport sciences, health sciences, and social sciences, and no multidisciplinary comprehensive research has yet been conducted (Johnson et al., 2017; Pantelaki et al., 2021). By reviewing multidisciplinary knowledge, this study analyzed the theoretical results of aging, social inclusion, and transport equity, discussed their theoretical contributions to the study on the mobility of disadvantaged traveler groups, explored the interaction mechanism between transport inclusion and social inclusion, and

established a theoretical analysis framework for elderly transport inclusion (Fig. 6).

3.1. Negative aging and active aging

Health sciences have contributed to a shift in the aging conceptualization from the negative to the active. Early aging research had a negative view, influenced by disengagement theory, which viewed aging as a decline in physical attributes and a gradual, universal and inevitable process of social role disengagement (Foster and Walker, 2014). Older people's social activities, social roles and meaning in life were framed within a limited context (Estes et al., 1992). With the birth of active psychology, the study on aging shifted to a constructive research stance. Active aging seeks to enable the elderly to realize their potential for physical, social, and spiritual well-being throughout the life course and to participate in society according to their needs, desires, and abilities, while providing them with adequate protection, safety, and care when they need. The active aging research framework

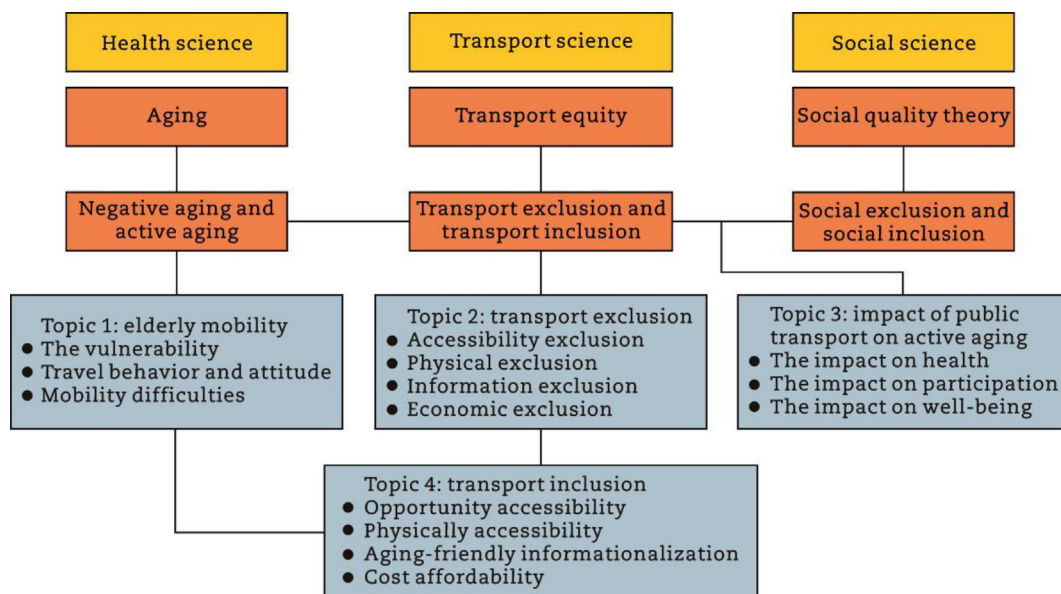


Fig. 6 – Review of research topics on elderly mobility.

converts the elderly from concentrators of social problems, consumers of social resources, and stressors on social development to social problem solvers, creators, and drivers for sustaining social development (Pérez-Cuevas et al., 2015).

A series of studies in the health sciences have demonstrated the impact of being out and about in social activities on the active elderly life status. Older people who continue to work have higher perceptions of life quality, and caring for their grandchildren provides them with higher feedback on both life quality and well-being (Ramia and Voicu, 2022). Volunteering outside the home prevents loneliness, increases self-esteem, expands social networks, enriches social capital, and promotes social integration (Saz-Gil et al., 2019). The International Council on Active aging (ICAA) strongly believes that “people can significantly improve the quality of their later years by staying active and participating fully in life” (ICAA, 2018). Currently, the concept of active aging is not only commonly utilized in policy, but is also valued by researchers in gerontology and the social sciences (Pfaller and Schweda, 2019). Gerontological studies have discussed in depth the impact from mobility on elderly health and have included transport security in the active aging indicator system (Li et al., 2020a, 2020b).

3.2. Social inclusion and social exclusion

Due to globalization, the growing division between rich and poor has drawn sociological attention to socially inclusion. Social inclusion and social exclusion aim to explore the positive or negative social impacts from resource distribution. Social inclusion is the process which ensures that people are protected from exclusion. People have access to opportunities and resources to participate fully in economic, social and cultural life, and enjoy a normal level of well-being (Council of the European Union, 2004). In contrast, social exclusion is caused by the unequal power relations or unequal distribution on resources that result in excluded subjects being denied equal access to some development opportunities and rights, with negative long-term effects on individual and societal well-being (Popay et al., 2008). While social exclusion focuses on the topic of marginalization from an individual's normal social associations, social inclusion aims to improve the conditions of participation in society, especially for disadvantaged people (Paraschiv et al., 2021), and to help excluded individuals to reintegrate into society.

Social inclusion is an important concept in the theoretical system of social quality, which together with economic security, social cohesion and social empowerment, constitute a component of social quality assessment. Social inclusion is also one of the goals for sustainable development, advocating a proactive approach to enable all people in society, including vulnerable groups, to have their basic needs met and to live in dignity. An important element of social inclusion is overcoming constraints on the travel ability to access social opportunities (Eager et al., 2023). At the individual level, social inclusion implies an ability to engage, to be able to visit friends, participate in social activities, work, or receive an education, and this ability to engage is often based on mobility (Stanley et al., 2011b). Social inclusion is associated with life satisfaction (Cantekin and Arpacı, 2023). On the

social level, social inclusion as a concept of sustainable urban development focuses on the combination of environmental, social and economic opportunities for disadvantaged groups (Mirzoev et al., 2021). Among them, spatial integration is an important step in the process of social inclusion (Massey and Mullan, 1984). Transport provides a condition that connects all types of development opportunities, and without the spatial integration of society by transport, the integration of social groups is difficult to occur.

Based on the interaction of transportation science and social science, the TRSE research topic has arisen. TRSE is a process whereby disadvantaged groups are inhibited in their ability to participate in daily activities through transport at a normal standard of living. This results in a lack of opportunity to participate in activities generally accepted by society. It is a process of social marginalization or segregation. TRSE research focuses on mobility deprivation of disadvantaged groups and its negative social externality effects (Lucas, 2019; McDonagh, 2006), examines the social differentiation caused by transport, discusses the impact from transport deprivation on their social opportunities, and works to identify transport exclusion. Focusing on TRSE as a major driver for social change (Button and Nijkamp, 1997), the application of the social exclusion concept to the transport inequality research not only opens up new research areas but also facilitates the development and application of new methodologies (Lucas and Jones, 2012). One study used the predicted social inclusion intensity to examine the causes of the decline in demand for air travel (Santos et al., 2021). An analysis based on the index of transport deprivation measure showed that transport accessibility deprivation accounted for 9%–10% of all types of deprivation measures, including English Indices of Deprivation (EID) 2007, Northern Ireland Multiple Deprivation Measure (NIMDM) 2005, Scottish Index of Multiple Deprivation (SIMD) 2006, and Welsh Index of Multiple Deprivation (WIMD) 2008 (DCLG, 2008; NISRA, 2005; Scottish Executive, 2006; Welsh Assembly Government, 2008). Mobility barriers produce exclusion in the social domains of income, unemployment, political engagement, social activity participation and social support networks (Currie and Delbosc, 2010).

3.3. Transport exclusion and transport inclusion

Barriers to mobility for vulnerable groups are a globally pervasive problem. As cities continue to expand, mobility for social members becomes less of an option and more of a necessity. The mobility dilemma disproportionately affects the poorest and most vulnerable. People are unable to participate in the processes of economic, political and social life in their communities, due to poor accessibility to development opportunities, services and socialization. There is a direct causal relationship between transport and social exclusion (Hernandez and Titheridge, 2016; Özkazanç and Sönmez, 2017). Mobility is directly related to the well-being level of an individual and has a significant impact on the social integration of social members. Further, the relationship between transport and social exclusion not only reflects the problem of mobility differentials and opportunity deprivation, but also reflects the dysfunctionality of social

relations between different social groups. In cities that are highly dependent on PT, vulnerable groups can become handicapped by mobility barriers.

In view of the increasing findings about the effects on social inclusion and social exclusion due to transport disadvantage, the transport study has begun to reflect on the one-sidedness of the cost-benefit analysis approach, which has dominated the evaluation on transport projects and policies for more than 50 years. The cost-benefit analysis approach is fundamentally oriented towards prioritizing the efficiency of urban development. It takes the increase in the general and average welfare in society as the main criterion for evaluating transport development, ignoring the impact on transport equity due to differences in social resources and capabilities. The disadvantaged traveler groups, although they can enjoy PT services equally, are limited by their travel capabilities, and they are in fact unable to obtain travel opportunities. If cost-benefit analysis is used as a criterion for evaluating transport policies, that means transport policies meeting the needs of the disadvantaged traveler groups but failing to meet the cost-benefit principle will not be implemented.

Transport is a national distributional domain. The sustainable national development requires a balance between economic efficiency and social equity in transport. The two logics of welfare distribution, “efficiency first” and “equity first”, belong to two different philosophical camps: utilitarianism and moralism. The two philosophies determine different starting points for transport policy. “Efficiency first” uses higher performance systems to address the mobility needs in society. For example, planning express buses, subways, and bus networks for peak commuter demand. Equity first addresses PT supply and availability for groups in mobility dilemma. It seeks to explore and utilize principles, rules, and institutions that contribute to the justice achievement, by deriving justice practices from policy implementation for equity goals (Zhou et al., 2019). Transport inclusion uses transport equity as the value orientation of the policy. John Rawls’ “a theory of justice” (Rawls, 1971) and Walzer’s “spheres of justice” (Walzer, 1983) provided the philosophical basis and distributional principles for transport inclusion. John Rawls’s justice theory has criticized utilitarianism, questioned whether free choices would necessarily bring about fairness and justice, and emphasized that the interests of socially disadvantaged groups should be prioritized in order to achieve equal opportunities. Spheres of justice, proposed by Walzer, has suggested that social goods that do not have unique significance could be dominated by the free value principle, but the goods in the public sphere should be subject to the distribution principle with a clear social value orientation. Specifically, transport is a public good with social significance that produces structured social effects on social groups. So the justice value of transport needs to be considered. According to the above two views on transport equity, transport should be an independent and non-market-driven distribution field. Therefore, it is necessary to conceptualize transport equity as a fundamental power paradigm rather than an individual responsibility (Fields et al., 2021).

Does equity then mean that parity must be achieved in the distribution of transport resources? From different

perspectives, the maximization criterion (Rawls, 1971), the hardship principle (Casal, 2007; Frankfurt, 1987), and the transport justice theory (Martens, 2012) have denied on this question. Rawls’ maximization criterion asserts that distribution is fair if it maximizes the prospects of disadvantaged groups within constraints, i.e., it should limit the maximum difference between the highest and lowest accessibility in terms of traffic patterns and traffic space, while maximizing average access. The hardship principle asserts not that everyone should have the same thing, but that everyone should have enough. That is, the distribution is fair if it ensures that everyone has the basic capabilities necessary for survival and personal development. The “transport justice theory” suggests that a comparative approach is more important than defining a perfect “end state”, i.e., focusing on ranking the injustices in social arrangements according to whether they are “more unjust” or “more just”.

The value shift towards transport equity has led many countries to work on identifying transport exclusion and translating the notion of transport inclusion into policy objectives. Transport exclusion defines those transport factors that bring social exclusion. Some scholars have used the terms transport disadvantage (Currie and Delbosc, 2010; Kamruzzaman et al., 2016; Stanley, 2004), transport poverty (Lucas et al., 2016), dimensions and indicators of TRSE (Yigitcanlar et al., 2019) and other similar concepts to describe transport exclusion (see the discussion in part 5.1 for more details). In contrast, transport inclusion is proposed as a more constructive concept. Transport inclusion is a transport policy framework, guided by the concept of transport equity, which focuses on access to equal mobility opportunities for disadvantaged traveler groups in order to eliminate the negative social effects caused by travel barriers to disadvantaged groups. Through constructing inclusive transport systems, policies and environments, support for the survival, development and well-being of socially disadvantaged groups can be realized. Inclusiveness is one of the principles of the WHO’s assessment framework for age-friendly cities, and the establishment of age-inclusive transport systems is essential to enhance their mobility and well-being (He et al., 2020). The vulnerability of public spaces in aging societies should be assessed to construct urban assessment models that take inclusive transport into account (Agost-Felip et al., 2021). The concept of inclusiveness should become a universally agreed upon street design paradigm (Asadi-Shekari et al., 2013; Wang et al., 2019). Transport services that are inclusive of nighttime travel needs should be provided for women (Plyushteva and Boussauw, 2020). Inclusive accessibility by proximity index (IAPI) was proposed in a study, showing that proximity is a valuable indicator of the ability to participate in social opportunities and networks (Lanza et al., 2023).

Given the importance of transport inclusion/exclusion for social inclusion and sustainable development, they have entered the policy agenda in many countries. In the United Kingdom, the Social Exclusion Unit (SEU) was established in 2003 to prioritize mobility and social exclusion on the transport policy agenda (Stanley et al., 2011a). Germany, Italy, and Japan have developed specific policies to address the

transport disadvantages of people with disabilities and older people, while U.S. policy emphasizes the concept of transport equity (Lucas, 2004). The European Commission funded a comprehensive review on transport projects to reduce TRSE in Europe (Holmes et al., 2007). Canada includes social equity and inclusion as part of national and provincial policy objectives (Allen and Farber, 2019), adhering to five values of action to improve social inclusion include: social justice, diversity assessment, opportunities for choice, rights and services, and cooperation (Shookner, 2002). China has made the “accessibility in the last mile” in poverty areas an important transport policy agenda for building a moderately prosperous society, and the country has included the equalization of PT services as a goal for sustainable transport development.

Interdisciplinarity and transdisciplinarity in regional planning research is seen to influence the processes and outcomes of PT and inclusive employment in urban planning policy (Tucker et al., 2022). Based on the literature review in this section, the life sciences provide evidence of elderly vulnerability, reveal the relationship between mobility and vital health, and promote a shift in aging theory from negative to positive policy goals, contributing knowledge to a deeper understanding of the vital status and needs of older people. Social inclusion and social exclusion provide theoretical concepts for understanding the indirect social effects from transport. Transport equity clarifies the ethical basis and resource distribution principles for transport and social opportunities for the inclusion of disadvantaged traveler groups. Based on the overlap between transport equity and social inclusion, the field of transport-related social exclusion (TRSE) research has emerged. Further, scholars have delved into the transport factors that generate social exclusion and have actively promoted policy practices for transport inclusion.

4. Elderly mobility and its social impact

4.1. Elderly characteristics

4.1.1. Elderly vulnerability

The health sciences have contributed a large body of research on the elderly, portraying their population characteristics. They are a particularly heterogeneous group (Fennel et al., 1993) with characteristics that distinguish them from other groups, representing a unique life stage. They are physically declining, have fragile lives (Prince et al., 2015), have a loss of independence (Shankar et al., 2017), and have higher security risks and safety concerns (Doi et al., 2016; Sochor, 2014). Age is a significant influence on self-reported stress and physiological data (Zhao et al., 2020). The elderly are more cost-conscious (Dominy and Kempson, 2006). Their ability to deal with problems is cognitively limited by previous life experiences (Kovalchick et al., 2004), and their ability to learn new technologies is declining (Selwyn, 2004). They value things that reflect their social values and are more pro-social (Van Lange et al., 1997), e.g., being helpful (Midlarsky, 1991), volunteering (Saz-Gil et al., 2019), being willing to be keeper of the meaning or taking responsibility

for future generations (Dávila and Díaz-Morales, 2009), and having a stronger sense of solidarity (Van Oorschot et al., 2005).

4.1.2. Elderly travel behavior and attitude

Research on elderly PT travel behaviors focuses on their PT choice, travel distance, and travel time. There are cohort differences (gender, age, disability, low income) in the travel patterns for older people (Mitra et al., 2021). Rural and urban areas also show different characteristics: the elderly walk most daily, use cabs occasionally or in emergencies, are highly dependent on cars, and PT is not a favored daily travel option for them (Graham et al., 2018). PT provision elements, including availability, reliability, accessibility, and affordability, affect their travel choices (Graham et al., 2018; Luiu and Tight, 2021). Individual factors also influence travel modes, including housing density, household vehicle ownership, driver's license ownership, age, gender, education, income, and employment status (Liu et al., 2020; Villena-Sanchez et al., 2022). In terms of walking distance, the elderly prefer shorter walking distances, and differences in walking preferences lead to significant heterogeneity in their assessment on PT accessibility (Liu et al., 2021c). The relationship between walking and shopping environments among older women is influenced by destination diversity and distance to supermarkets, while these factors have no effect on older men (Morioka et al., 2023). In terms of travel time, the elderly with childcare responsibilities travel in the morning peak as a matter of necessity, which leads to the limited attractiveness of free PT policies (Zhang et al., 2019). Older adults who do not have work and childcare responsibilities are more likely to stagger their trips (Rye, 2008). The elderly returning to work remain sensitive to travel times (Liu et al., 2020).

Research on elderly attitudes toward transport has focused on both their satisfaction and the social factors affecting their attitudes. In terms of travel satisfaction, existing studies have analyzed generational differences, between-group differences, and factors affecting travel satisfaction. One generational study shows that travel satisfaction increases linearly with age (Olsson et al., 2020). The effect of transport accessibility on travel satisfaction and life quality is more pronounced in younger older adults, but for those over 80 years of age, travel satisfaction decreases significantly (Kim et al., 2020). Factors affecting elderly travel satisfaction are accessibility (Kim et al., 2020), travel habits in younger years (Ingvardson and Nielsen, 2019), travel modes, travel time, transport costs, geographic location, socio-economic status, chronic health conditions, and social support (Li et al., 2020a, 2020b). The elderly, whose travel habits in early life affect their travel behavior in later life, have higher satisfaction with PT use than younger adults (Ingvardson and Nielsen, 2019). The elderly who walk to reach primary care have higher levels of satisfaction than those driving (Li et al., 2020a, 2020b). In addition, several studies have analyzed social factors that influence elderly attitudes towards transport. These factors include: social norms, pro-social values (Nikitas et al., 2018), social support (Peek et al., 2014; Stanley et al., 2011a), and the influence from underlying cultural norms. The underlying cultural norms

include the household registration system (Zhang and Zhao, 2021), transport spatial concession rights, e.g., preferential seating, cars yielding to people (Jones et al., 2013), and cultural impediments (Zhang et al., 2020). Rural older men perceive the freedom to travel as a symbol for independent identity (Graham et al., 2018). Compared to men, older women are less likely to hold a driver's license. They feel insecure about hitchhiking (Luiu and Tight, 2021). They view PT as a valuable social space (Graham et al., 2018). These established perceptual constraints affect elderly attitudes to travel and to transport policy.

4.1.3. Elderly mobility dilemma

Mobility dilemma is a travel barrier to the elderly caused by insufficiently inclusive transport services. Older people's personal driving ability declines with age (Ang et al., 2019), and once they stop driving they are rarely able to meet their travel needs (Luiu et al., 2017). Their physical limitations affect mobility patterns (Ang et al., 2019; Hidayati et al., 2021). Data from a study in the United States suggests that 16% of older adults' travel plans are not realized, potentially causing them to lose health opportunities (Fields et al., 2021). Another study reviewed 29 research articles for the issue of the unmet travel needs among the older population and showed that at least one-third of older adults reported their unmet travel needs. The most affected groups were female older adults and those aged 75 and over (Luiu et al., 2017).

Barriers to PT use among the elderly arise mainly from PT accessibility, physical accessibility, information barriers, and cost burden. First, for PT accessibility, the elderly face barriers in terms of connections, destinations, and access to transport services. Specifically, there is a lack of connectivity and networking between transit transfers or with other travel modes (Buys et al., 2012). Service delivery is inappropriate or unavailable in specific areas or to specific destinations (Su and Bell, 2009). The elderly face unreliable PT services, excessive distances to transit stops, long waiting time (Vine et al., 2012). Second, in terms of physical accessibility, the elderly experience difficulty getting on and off buses, insufficient senior priority seating, poor bus design, poor communication with staff (Accedes-Gonzalez et al., 2015; Ramachandran and D'souza, 2015; Windle and Burholt, 2003), the risk of falls from overcrowding (Zhang et al., 2019), and ageism from staff and other passengers (Ma et al., 2022; Neves et al., 2023; Nordbakke and Schwanen, 2014). Third, in terms of information barriers, the elderly have wayfinding/navigation difficulties and are excluded from digital travel with smartphones. They have difficulty understanding timetable and route information (Vine et al., 2012). They have significantly lower ease of use, perceived accessibility and perceived fairness in smartphone travel than the other population groups (Liu et al., 2021a). New technologies challenge the learning ability of older adults, who are not comfortable using smartphone-based services (Shirgaokar, 2020) and cannot access e-travel information provided through mobile devices (Mackett and Thoreau, 2015). Due to a lack of knowledge about location-based service (LBS) on smart phone, people over age 50 are often limited to traditional

sources of transport information (Brandtzæg et al., 2011; Zhang et al., 2020). Their caution about privacy also reduces their willingness to use ride-hailing booking technology (Misra et al., 2022). Fourth, older adults' sensitivity to transport costs affects their travel. They are more cost-conscious than younger people, with cost being a variable that influences their travel behavior (Sharaby and Shiftan, 2012). Income not only affects their ability to pay for road costs, but also their beliefs about the affordability for transport costs. Perceived value of pensions influences their receptivity to toll policies (Nikitas et al., 2018). When the travel cost increases by 1%, the probability of older adults choosing PT will decrease by 0.29% (Zhang et al., 2019).

4.2. Effect of elderly mobility on active aging

Based on social spillover theory, scholars have focused on how transport affects elderly lives. There is a positive correlation between mobility and active aging. The elderly whose mobility is met have better health, independence, and connectivity to society (Pantelaki et al., 2021). This is a consensus across countries on this issue. The active aging policy systems of China, Australia, the European Union, Russia, and Finland, among others, explicitly mention the important role of transport security (Li et al., 2020a, 2020b). Existing studies have discussed the relationship between transport and active aging mainly from three aspects: physical and mental health, social participation, and well-being. Active aging aims to develop health and social participation opportunities for the elderly to enhance their well-being. Inclusive transport environments are a prerequisite for them to achieve active aging status. Existing research provides critical support for understanding the relationship between transport inclusion and active aging.

4.2.1. Effect of PT on health

In the study of the relationship between transport and health, existing research focuses on the negative impact of mobility dilemma on older people's physical and mental health, and finds multiple lines of evidence that transport inclusion supports their health. Health science research has focused on how mobility affects elderly life quality. Mobility limitations increase falls (Dai et al., 2012; Musich et al., 2018), mortality (Ensrud et al., 2016; Mackey et al., 2016), cognition (Bishop et al., 2016; Demnitz et al., 2018; Rajtar-Zembaty et al., 2019), loneliness (Zeitler and Buys, 2015), depression (Jackson et al., 2019), obesity (Webb et al., 2012), and disability risks as well (Pantelaki et al., 2021). Transport research indicates that travel can provide multiple supports for elderly health and is vital for their physical and psychological well-being (Ziegler and Schwanen, 2011). Differences in access to PT may exacerbate health and social inequalities (Franke et al., 2019). More frequent PT use is associated with better physical condition and perceived psychological self-perceived health (Crotti et al., 2021). Bus passes encourage more physical activities to reduce obesity (Van Cauwenberg et al., 2018; Webb et al., 2016). Inclusive transport can significantly improve self-reported health and mental stress among the elderly (Dharmowijoyo et al., 2020; Liu et al., 2019). Traveling by bus can provide cognitive stimulation

with positive effects on cognitive function improvement (Cassarino and Setti, 2016; Mokhtarian, 2005; Reinhard et al., 2019), especially in terms of increasing memory scores (Reinhard et al., 2019). Better PT accessibility is associated with lower levels of mental distress (Cheng et al., 2016). Transport accessibility affects healthcare access (Li et al., 2020a, 2020b; Yuan et al., 2019). Travel satisfaction affects health (Lucas, 2012) and life satisfaction (Gao et al., 2018).

4.2.2. Effect of PT on social participation

In research on the relationship between transport and social participation, existing studies have focused on how transport can support the social inclusion and the realization of social roles for the elderly. Specifically, there are three aspects. First, existing research reveals the impact by PT on older people's participation in social activities and their access to social opportunities. Low social participation is associated with low mobility (Raymond et al., 2014; Rosso et al., 2013). Among the elderly who have lost their ability to drive, those who are better able to participate in PT travel modes have higher levels of social participation, quality of life, and social cohesion (Feng, 2017; He et al., 2018; Voss et al., 2016). Improving transit accessibility for low-income and careless households would effectively increase their activity participation (Allen and Farber, 2020). Especially for older adults with mobility impairments, social participation significantly increases their well-being (Siren et al., 2015). Second, existing research has revealed the impact of PT on elderly social inclusion. Among the elderly in transport poverty, the most apparent issues are the shrinkage of social networks and the collapse of a sense of belonging (Currie et al., 2010; Levasseur et al., 2015; Lucas, 2012). Social isolation is linked to increased mortality and functional decline in the elderly (Brandt et al., 2022; Holt-Lunstad et al., 2015; Shankar et al., 2017). In rural areas, transport systems can help older adults move out of isolation, maintain autonomy, and provide informal spaces for their interactions to build social relationships and access local news (Hagan, 2019). A good transport system supports them to engage in community activities, expand social interactions, and rebuild their sense of community (He et al., 2018; Hemingway et al., 2013). Older people's transport satisfaction is closely related to their sense of community, and they feel socially integrated when the transport system works well (He et al., 2020). Third, existing research has revealed the value of transport for the elderly to return to productive roles. Elderly mobility and their ability to live independently have positive implications for the development of an aging labor resource. The elderly can gain life satisfaction and social fulfillment through productive activities outside their homes, such as volunteering, returning to the workplace, or caring for others (Friman et al., 2017; Nordbakke and Schwanen, 2014).

4.2.3. Impact by PT on well-being

In the research on the relationship between transport and well-being, existing studies have revealed two mechanisms by which transport affects elderly well-being. Specifically, inclusive transport can help them reintegrate into social roles to enhance well-being, or improve well-being by increasing their

travel satisfaction. The findings of these two aspects are as follows. First, inclusive PT can encourage older people's reintegration into society, thus increasing their well-being. Upon retirement, the elderly experience a fading of their original social roles, which used to provide them with purpose and social connections in their lives. Mobility can help them meet psychosocial needs, which are necessary for good well-being (Vella-Brodrick and Stanley, 2013). Inclusive PT promotes elderly social participation, which can help them maintain their social roles and seek meaning in their lives, thus increasing their well-being (Heaven et al., 2013). Higher travel rates are directly associated with social belonging and indirectly enhance well-being (Stanley et al., 2011a). Conversely, transport disadvantages (economic cost to travel and option limitations) affect travelers' perceptions to freedom, leading to differences in subjective well-being (De Vos et al., 2013). Transport disadvantage is positively correlated with social exclusion, and the social exclusion intensity is strongly negatively correlated with well-being (Currie and Delbosc, 2010). Second, according to the bottom-up spillover theory, travel satisfaction can spill over into the life domain, thereby increasing life satisfaction. Therefore, travel satisfaction has a positive impact on the social inclusion and well-being of the elderly. For example, one study discussed the mechanisms by which travel satisfaction affects life well-being. Daily travel affects travelers' moods, leading to cumulative satisfaction, which in turn directly or indirectly affects subjective well-being (Bergstad et al., 2011). A study, combining the psychological scale of subjective well-being with the travel satisfaction scale, revealed the effects of positive emotions, negative emotions, and cognitive appraisals on subjective well-being (Singleton, 2019). One study found that older people's satisfaction with the transport system interacted with social inclusion and also positively affected their physical well-being and psychological well-being (He et al., 2020). Another study also supported this view. It found that satisfaction with the transport system influenced social inclusion, which was a potential mediator between perceived transport system and well-being (He et al., 2020).

Furthermore, in addition to physical and mental health, social participation, and sense of well-being, existing research has also identified a number of other valuable variables in the relationship between transport and aging, such as psychosocial barriers, community environment, social capital, and family factors. This is revealing for a more comprehensive understanding of the relationship between transport and elderly social exclusion. These research models and variables have opened up new research spaces, which can help explore the relationship between transport and aging at the family and social levels. For example, using the integrated behavioral model, one study found a significant association between psychosocial barriers and social exclusion in elderly travels by PT. These psychosocial barriers include perceived norms, attitude towards PT, perceived control and self-efficacy (Al-Rashid et al., 2020). Other non-psychological barriers (e.g., neighborhood social environmental constraints and habits towards PT travel) also affect social exclusion (Al-Rashid et al., 2021). A study explored the effects from neighborhood environment and transport on subjective well-being. It

found that neighborhood environment and community cohesion have a positive effect, while neighborhood density and perceived crime in the neighborhood have a negative effect (Ma et al., 2018). Based on self-determination theory and mental health model, a study found the role of social capital, community extraversion, household income, and trips on social exclusion (Stanley et al., 2011a). In addition, based on the theory of planned behavior, one study revealed a negative correlation between symbolic value for private cars and bus use (Li et al., 2019).

Overall, the research on PT and active aging belongs to category studies. The studies focus on travel characteristics, mobility dilemmas, and transport-related social exclusion/inclusion for specific disadvantaged traveler groups. In such studies, self-determination theory, behavioral theory, and social spillover theory are more often applied to explain the impact of travel behaviors or attitudes on the quality of elderly life. Data for this type of research are mainly derived from subjectively reported resource, such as questionnaires (Luiu and Tight, 2021). The data are more often analyzed using statistical methods such as logistic regression (Luiu and Tight, 2021), SEM (Currie and Delbosc, 2010), PLS-SEM (Al-Rashid et al., 2021; Yuan et al., 2019), multivariable analysis, multiple imputation techniques and sensitivity analysis (Di Gessa and Grundy, 2014). Qualitative research methods have also been applied, such as focus groups interviews and in-depth interviews, to explore the inclusion on new sociological variables in the research model. According to the literature review, the spillover effects from elderly mobility barriers were identified in the areas of healthcare resource accessibility (Li et al., 2020a, 2020b), physical and mental health (Dharmowijoyo et al., 2020), social participation (Levasseur et al., 2015), life satisfaction (Olsson et al., 2020), social isolation (Shankar et al., 2017), and happiness (Siren et al., 2015). These contribute valuable knowledge for researchers to understand the indirect social effects from transport, demonstrating the importance of transport equity as a principle for transport resource distribution and policy evaluation. However, such studies rarely consider detailed geographic factors, origins and destinations, and travel capability differences (Church et al., 2000), posing limitations to the equitable transport resource distribution and also transport inclusion policy practices. Therefore, there is a need to identify transport exclusion factors affecting elderly mobility and to develop a research framework for PT inclusion.

5. A research framework for elderly PT exclusion/inclusion

Based on the theories related to transport inclusion and the literature review on transport and active aging, this section further comprehends the transport exclusion factors affecting elderly mobility. On this ground, this study proposes a research framework for elderly PT inclusion in terms of opportunity accessibility, physical accessibility, age-friendly informationalization, and cost affordability.

5.1. Identification of elderly PT exclusion factors

As can be seen through Subsection 4.1, the elderly are experiencing a fragile life stage, with travel characteristics distinct from other age groups, and they face mobility barriers to travel by PT. Transport exclusion and transport inclusion affect elderly mobility and have a negative or positive impact on aging. For this reason, Subsection 4.2 analyzes the impact by PT on the positive aging goals for health, social participation, and well-being. In order to clarify the factors causing elderly mobility dilemma, scholars have developed a theoretical analytical framework through a number of academic concepts, including transport disadvantage (Currie and Delbosc, 2010; Kamruzzaman et al., 2016; Stanley, 2004), transport poverty (Lucas et al., 2016), factors that may limit the mobility of socially excluded people (Church et al., 2000), dimensions and indicators of TRSE (Yigitcanlar et al., 2019) and functional requirements for inclusive transport (Bjerkan and Øvstedal, 2018).

In these similar academic concepts, transport disadvantage is defined as people's limited mobility due to a lack of transport options, thus limiting their access to goods, services and relationships (Stanley, 2004). "Transport poverty" is generally used to describe situations when people do not have access to essential services or work because of a lack of affordable or available transportation options" (Kiss, 2022). These concepts can be seen as multi-dimensional and there is still no agreed definition (Verhorst et al., 2023). Given the conflation of different terms and overlapping meanings, scholars have tended to create a lexicon of definitions to ensure greater conceptual clarity and consistency in academic and policy discussions (Lucas and Markovich, 2011). For example, Lucas et al. (2016) uses a way of combined subset to describe transport poverty.

Regardless of whether there is a concise conceptual expression, the proximate concepts for transport exclusion mentioned above proposed transport factors that contribute to the social exclusion, such as accessibility exclusion, physical exclusion, information exclusion, economic exclusion, and so on. There are some subtle differences among these concepts (Table 7). Currie and Delbosc (2010) developed the definition of "transport disadvantage" by adding non-transport environmental factors ("vulnerable/impaired", "rely on others"). Based on a review on the concepts of "transport/mobility poverty" and "accessibility poverty", Lucas et al. (2016) proposed a definition of "exposure to transport externalities". Gandelman et al. (2019) added "time poverty" to the conceptual framework of "transport poverty" based on the work of Lucas. Compared with the concepts of "transport disadvantage" and "transport poverty", the indicators of transport exclusion are more detailed and specific. In particular, recent studies have added the information dimension (Bjerkan and Øvstedal, 2018; Yigitcanlar et al., 2019), which is more relevant for understanding elderly transport exclusion in the digital transport environment. In addition, because the research goal for this paper is to analyze the transport factors affecting elderly mobility and contributing to TRSE, without considering the personal and

Table 7 – Detailed information for the concept of TRSE.

Reference	Study question	Location	Data	Method	Term & indicator	Common				Difference
						Accessibility exclusion	Physical exclusion	Information exclusion	Economy exclusion	
Kamruzzaman et al. (2016)	The paper assessed effectiveness using criteria derived through a review of the concepts of transport disadvantage and social exclusion and their operationalization.	Unmentioned	155 pieces of literature	Review	Transport disadvantages (1) Disaggregation of socio-demographics and socio-economic (2) Spatial accessibility: accessibility of different types of opportunities (3) Temporal accessibility: accessibility of opportunities temporally (4) Spatial mobility: accessibility of transport spatially (5) Temporal mobility: accessibility of transport temporally (6) Interaction: interactions between different causal factors (7) Relativity: activities of others living in the same area	√	√		√	Non-transport factors, e.g., socio-demographics and activity space, were taken into account.
Currie and Delbosc (2010), Delbosc and Currie (2011)	This research modeled the social and psychological impacts of transport disadvantage.	Melbourne, Australia	535 household interview surveys	(1) Principal (2) Component (3) Analysis/structural equation modeling	Transport disadvantages (1) Transit disadvantage: operation time, frequency and connection (2) Transport disadvantage: destination accessibility, reliability, convenient site, fast arrival to the destination, reasonable time (3) Vulnerable/impaired: physical impairment, cognitive impairment, and security (4) Rely on others: need help from others or bear transport costs during travel	√	√		√	Non-transport factors, e.g., vulnerability of travelers and social support, were taken into account.
Lucas et al. (2016)	Demonstrating how the different ways that academic studies and policy programmers have defined and recorded the problem of transport poverty is directly related to the practice in developed countries and developing countries.	Lack of detailed instructions	National Travel Survey (NTS)/EU Statistics on Income and Living Conditions (EU-SILC)	Review of existing studies and analyze the merits and limitations of available data in Britain	Transport poverties (1) Mobility poverty: lack of services or infrastructures (2) Accessibility poverty: lack of reasonable time, ease and cost to reach key activities (3) Transport affordability: lack of individual/household resources to afford transport (4) Exposure to transport externalities: road traffic casualties and chronic diseases and deaths from traffic related pollution	√	√		√	Non-transport factors were taken into account, e.g., exposure to transport externalities.

(continued on next page)

Table 7 – (continued)

Reference	Study question	Location	Data	Method	Term & indicator	Common				Difference
						Accessibility exclusion	Physical exclusion	Information exclusion	Economy exclusion	
Gandelman et al. (2019)	This paper described household spending on transport in Latin America and the Caribbean and defined transport poverty.	12 Latin America countries and the Caribbean countries: Bahamas, Bolivia, Brazil, Chile, Costa Rica, Ecuador, Honduras, Mexico, Nicaragua, Panama, Paraguay, and Uruguay.	(1) Date form: Engel curves diaries and interviews (2) Date time: different years from 2003 to 2014	Engel curves methodology	Transport poverties (1) Transport mode option: no transport option available to individual's physical condition an capabilities (2) Destinations accessibility: existing transport options do not reach key destinations (3) Cost affordable: weekly amount spent on transports leaves the household with a residual income below the official income line (4) Time poverty: need to spend an excessive amount of time traveling, leading to time poverty (5) Unsafe travel conditions: the prevailing travel conditions are dangerous, unsafe or unhealthy for the individuals	✓	✓		✓	Non-transport factors were taken into account, e.g., exposure to transport externalities; transport factors were added, e.g., time poverty.
Church et al. (2000)	The study proposed a conceptual framework of TRSE and examine a selection of indicators which might be used in assessing the outcomes of policies designed to use increased mobility to reduce exclusion.	UK	An example of the use of London Transport's Capital model	Review of current practice and research	Factors of TRSE (1) Physical exclusion: transport environment barrier-free (2) Geographical exclusion: poor accessibility to destinations (3) Exclusion from facilities: distance leads to money and time costs increase (4) Economic exclusion: travel costs can limit the geographic range of social opportunity (5) Time-based exclusion: the effect of time poverty (6) Fear-based exclusion: lack of security (7) Space exclusion: the effect of security and space management policies	✓	✓		✓	Transport factors were added, e.g., security and space management.
Yigitcanlar et al. (2019)	The paper identified dimensions and indicators of TRSE through the review and assessed their suitability by	Online questionnaire	33 international experts	Review/delphi survey	Factors of TRSE (1) Physical condition (2) Economic: income, public transport cost, private transport cost (3) Temporal: service frequency or waiting time, transport availability, travel time	✓	✓	✓	✓	Transport factors are added, e.g., information exclusion.

	international experts.				(4) Spatial: distance to bus stop/ferry terminal/train/light rail, distance between transfer points (5) Psychological: area accessibility level, safety measures (6) Information: security, information use, information availability					
Bjerkan and Øvstedal (2018)	This study summarized functional requirements for inclusive transport of disabled people.	Unmentioned	34 empirical studies	Review	Functional requirements for inclusive transport (1) Information: accessible, understandable and identifiable (2) Flexibility: diversity and flexibility of travel options (3) Safety and security: feeling physical safety and psychological environment friendliness (4) Physically accessible design: barrier-free facilities (5) Reliability: trust and predictability of the service (6) Economic: predictability (7) Administration: reduced administration (8) Travel times: short and predictable	√	√	√	√	Accessibility factors were ignored, and management factors were increased.

social factors that cause mobility barriers, the concept of transport exclusion is therefore more accurate.

In this paper, “transport exclusion” is defined as four transport factors that cause TRSE, including opportunity accessibility exclusion, physical accessibility exclusion, information exclusion, and economy exclusion. (1) Opportunity accessibility exclusion is a dimension in which the elderly are inhibited by temporal accessibility and spatial accessibility. This dimension takes into account location, distance, and time. (2) Physical accessibility exclusion is when the elderly have difficulty using transport facilities or have safety issues due to physical and cognitive decline. This dimension is related to the accessibility level of transport facilities and spaces. (3) Information exclusion is when the elderly have difficulties in accessing, recognizing, and perceiving transport information. This dimension is related to the information divide. (4) Economy exclusion is that the elderly are inhibited from traveling because of transport cost pressures. This dimension is related to the economic affordability of older adults.

5.2. Research framework for elderly PT inclusion

5.2.1. Opportunity accessibility

Transport is closely related to the distribution of life opportunities, and accessibility is often used in urban transport assessments as a key indicator of the rationality in transport facility distribution. Transport accessibility is defined as the “potential for interaction opportunities” and measures the number and diversity of opportunities to reach activities, services and goods through the transport system (Hansen, 1959; Martens, 2012). For the elderly, PT accessibility focuses on destinations such as health, financial services, cultural, leisure, and commercial services (Gaglione et al., 2021; Rojas et al., 2016), with a particular focus on healthcare (Cheng et al., 2020; Li et al., 2020a, 2020b) and parks (Zhang et al., 2022). How PT accessibility is equitably distributed among social members to achieve equality in social opportunities is a key factor in assessing the quality of life for the elderly (Musselwhite and Haddad, 2010).

Accessibility can reflect the supply-demand ratio relationship between two points within a critical range, which can be used to guide the PT siting and planning. The accessibility evaluation usually addresses the following four categories (Lei and Church, 2010; Malekzadeh and Chung, 2020; Wang and Chen, 2015). System accessibility uses the cumulative method to determine the population coverage within the radius of a station, or the shortest distance method to measure the distance from a neighborhood to a transit station. System facilitated accessibility is measured using the OD impedance matrix, distributing gravity based on location and calculating the travel time cost from neighborhoods with different socio-demographic attributes to the destination area. Integrated accessibility is measured using a gravity-based or two-step floating catchment area (ASFCA) to measure the number of opportunities that can be reached by the transit system within a certain time or distance range. Relative accessibility examines the accessibility of PT trips in terms of travel time and cost compared to other travel modes, and allowing for the

assessment on travel modal mismatches in different neighborhoods.

As walking is an integral part in the PT travel chain, several studies have included walking in accessibility models for the elderly. One study by introducing street width and slope, analyzed the walking time for elderly people to access the street network of PT facilities, and found differences in accessibility across age groups (Achuthan et al., 2010). A study considered the walking and PT networks, the location of basic urban services, and the organization of urban resources. By using the 2SFCA, it proposed a measure of accessibility to healthcare services. It showed that nearly 40% of older adults had poor access to primary healthcare services (Guida et al., 2022). Other studies have focused on spatial variation in walking accessibility among the elderly, to identify areas at TRSE risks. For example, one study proposed a new methodology for assessing the walking accessibility for the elderly in different types of cities, according to their walking speed, pedestrian network characteristics, pedestrian safety features, and the pleasantness and attractiveness of the walking environment (Gaglione et al., 2021). A study modeled the accessibility of whole-trip traveling for elderly travel barrier weights and barrier probabilities (Sundling et al., 2015).

Widely used accessibility measurement techniques include (Table 8): distance-based models, cumulative models, gravity-based models, two-step floating catchment area models (ASFCA), utility-based models, and social network analysis (Guagliardo, 2004; Hu et al., 2013; Mavoa et al., 2012). These measurement techniques can evaluate accessibility respectively in terms of spatio-temporal relationships, impedance factors, supply-demand relationships, transportation networks, and utility maximization. Different methods have their own application space. Appropriate methods should be selected according to the problem context and data conditions. Without combining the actual research problem context, it is impossible to determine which accessibility measurement method is optimal (Van Wee, 2016).

Spatial category analysis approach is usually applied for the study of accessibility and transport justice. It aims to study spatial differentiation in accessibility to identify areas where transport generates social exclusion. Variation in PT accessibility will lead to differences in the spatial distribution in educational opportunities (Tiznado-Aitken et al., 2021), job opportunities (Chaloux et al., 2019), and amenities (Church et al., 2000). Spatial category studies require transport data along with demographic data to reflect spatial differentiation in accessibility by calculating supply-demand differences and inequality indices. Inequality indices (e.g., Theil index, Gini coefficient, Atkinson index, Kolm index, etc.) are often used in transport accessibility and equity studies to analyze the fairness in transport resource distribution to low-income groups. Existing studies have shown that there are limitations in applying a single index, e.g., the same Gini coefficient may represent different accessibility distributions (Cheng et al., 2020). Therefore, it is recommended to use multiple inequality indices to avoid the shortcomings of a single index (Ramjerdi, 2006).

The rise of such technologies as transport digitalization and big data analytics has provided big data support for spatial

Table 8 – Comparison of accessibility measurement techniques.

Measurement technique	Feature	Advantage	Disadvantage	Reference	Other note
Distance-based model	Incorporates distances from a given origin to different opportunities into the accessibility model.	Simple and easy to implement: relatively little data, with results easy to understand, use, and interpret.	Adoption of fixed-distance ratios poorly adapted to different cases.	Malekzadeh and Chung (2020)	
Cumulative models	Explains accessibility by measuring the cumulative number of potential opportunities available within a given time or distance threshold.	With relatively little data, with results easy to understand, use, and interpret.	Predominantly measured by objective indicators, making it difficult to comprehensively measure the ease with which older people can acquire accessibility opportunities.	Chen et al. (2022) , Deboosere and El-Geneidy (2018)	Differences in accessibility across transport modes and communities can be assessed through the weighting of relative accessibility and the coefficient of variance (Chen et al., 2022).
Gravity-based models	Measures the potential opportunities for interaction by impedance functions, which are travel distance, travel time, and travel cost.	With the scale of supply and demand and various impedance factors considered, widely used.	Complex and not easy to understand; with the multimodal nature of travel behavior difficult for gravity models to accurately capture attenuation functions and trade-off opportunities.	Batty (1976) , Bertolini et al. (2005) , Bocarejo and Oviedo (2012) , Chen et al. (2018) , O'Sullivan et al. (2000)	To address the shortcomings of fixed thresholds, a study proposed adaptive thresholds to measure reachability, which could improve behavioral realism (Cheng et al., 2019).
Two-step floating catchment area models, ASFCA	Uses the principle of gravity modeling and searches twice based on the supply and demand points moving within a certain range of thresholds, to determine the accessibility of the demand points by adding the supply and demand ratios.	With supply and demand scales, and critical distances considered; widely used, with several improved models.	With insufficient consideration of the role of distance attenuation; with more data requirements for model calibration; ignorant of spatial interactions and potential competition between supply and demand points.	Liang et al. (2023) , Radke and Mu (2000)	Improved models include: Gaussian-based two-step floating catchment area method (G2SFCA) (Dai, 2011); multi-mode Gaussian-based two-step floating catchment area method (G2SFCA) (Hu et al., 2020); improved three-step floating catchment area method (3SFCA) (Liang et al., 2023); dynamic two-step floating catchment area method (D2SFCA) (Wu and Zheng, 2023); age-grouped Gaussian-based two-step floating catchment area method (AG2SFCA) (Li and Wang, 2024); kernel density two-step floating catchment area method (KD2SFCA) (Zheng et al., 2020); Huff Gaussian-based 2SFCA (H-G2SFCA) (Zhang et al., 2021).

(continued on next page)

Table 8 – (continued)

Measurement technique	Feature	Advantage	Disadvantage	Reference	Other note
Social network analysis	Evaluates the ability of travelers to access remote opportunities through a particular station.	With quantitative evaluation of transport connectivity, with analysis of the degree of convenience of each station in terms of access to buses, transfers and destinations, and measurement of the importance, independence, proximity, and influence of PT stops.	Focuses on the structural pattern of the transport network, which is static and lacks dynamic analysis.	Kaplan et al. (2014), Lanza et al. (2023), Sharma et al. (2020), Welch and Mishra (2013)	
Utility-based models	Adopts the logit-based utility model, the principle of maximizing the traveler's utility is incorporated into the accessibility evaluation.	With behavioral characteristics of decision makers maximizing utility considered, reflecting the subjective choices of travelers.	With only travelers' site or destination accessed behavior incorporated unable to explain travelers' choices or perceptions of the transport network.	Kelobonye et al. (2019)	Examples can be: the environmental transit accessibility index (ETAI), which considers the random utility of accessing transit stations (Rastogi and Rao, 2003); and the LUPTAI model, which considers destination preferences (Davidson, 2008).

category studies, stimulating an increase in the number of spatial differentiation studies. Big data methods have advantages in capturing the spatial characteristics in objective transport environments and the spatio-temporal details on travel behavior. Transport digitalization can provide online maps, GPS transit operation data, PT card data, and user cell phone signaling data, which are accurate and real-time research data. Based on these, spatial category analysis on mobile activities can be performed using computational methods of GIS. Spatial distribution of transport poverty can be described using cluster analysis (e.g., k-means) (Allen and Farber, 2019). However, existing studies on big data-driven spatial differentiation have trouble in mismatches in the sampling period of multi-source datasets, insufficient granularity of demographic data, limitations in proxy indicators (e.g., property prices as a proxy for the economic status of a community do not accurately account for a mixed-rich-poor neighborhood) (Chen and Yeh, 2019), and as well the personal privacy risks associated with big data. In particular, vulnerable groups that require special attention in terms of transport inclusion are also often the disadvantaged in information technology, and they are often excluded from big data (e.g., the elderly, the poor, children). Bias in the assessment on information disadvantaged groups by big data technologies can affect the accuracy in transport inclusion evaluations. Moreover, based on national geographic security and personal data privacy considerations, access to these data is severely restricted, which poses challenges for research. Overall, despite the many advantages of big data-driven spatial equity research, the existing research results are more theoretically repetitive in verifying the geospatial differences in transport equity, and breakthroughs are still needed in discovering new knowledge and theories on the production mechanism of transport disadvantages (Zhang and Zhao, 2021).

5.2.2. Physical accessibility

The elderly are a vulnerable group of citizens, facing declining vision, hearing, cognition, and mobility. They experience difficulties in traveling to and from stations and getting on and off vehicles. In response to elderly mobility disadvantages, existing studies have examined the physical accessibility to transport in terms of inclusive PT, inclusive streets, and paratransit connections.

The physical space design for inclusive PT needs to consider those with physical disabilities, who travel with white-tipped canes, assistance dogs or wheelchairs, etc. One study reviewed barriers faced by individuals with disabilities when using regular transit networks and made inclusive recommendations of universal design (Table 9). Another study reviewed the design standards and requirements in the accessibility guidelines in the United States, the United Kingdom, and Hong Kong, China. These design standards specify the minimum required dimensions of building areas (including public access routes, ramps, doors and/or entrances, stairs, escalator, lifts), transport facilities (including signage/information, bus stops, railway platforms and railway stations), traffic control facilities (curbs, pedestrian crossings, footbridge and/or underpass), and transport vehicles (including details on bus vehicles, railway

Table 9 – Inclusive environmental design for PT (Garnier et al., 2022).

Composition	Detail
Vehicles (buses, metro, train, etc.)	1. Presence of a functional bus ramp, lift and kneelers 2. Use of anchor straps in buses 3. Reserved seats 4. Reserved spaces 5. Availability of stop announcement
Stations	1. Presence of escalators 2. Presence of automatic doors 3. Presence of elevators 4. Installation of adequate lighting 5. Installation of air conditioning
Bus stops	Presence of bus shelters
Platform	1. Reserved seats 2. Proper gaps between platforms and rail cars
Indications and support	1. Availability of human support 2. Presence of trained staff 3. Availability of accompanist 4. Presence of braille, visual and audible signage, and alarm

rail vehicles) (Sze and Christensen, 2017). One literature used a point system to evaluate inclusive walking facilities, with evaluation metrics including: slopes, elevators, curb ramps, wheelchair-accessible drinking fountains, tactile pavement, ramps, toilets, garden and signal (Agost-Felip et al., 2021).

The design on roadway environments is critical to elderly transport safety and travel experience. Data show that the number of elderly transport fatalities has risen with the rapid growth of a super-aged population in Japan. The elderly account for 55% of all transport fatalities, 19% of walking accidents, and 76% of crossing accidents (Doi et al., 2016). In response to road traffic safety, a study proposed a governance framework for transitioning to safer streets by integrating the unconscious habits by pedestrians, the road space compactness, and the priorities of transport participants (Doi et al., 2016). Older people's sensitivity to walkable environments makes them resonate more strongly with inclusive streets (Doi et al., 2016). One study proposed six dimensions for perceived inclusive streets: accessibility, connectivity, streetscape, safety and security, urban vitality, and greenery (Wang et al., 2019). A study proposed a standard for transport technology to improve elderly accessibility, which covered aspects such as greenery, noise, ramps, pavements, pedestrian routes, pavement maintenance, cleanliness, accessible transportation, and traffic lighting (Agost-Felip et al., 2021).

Paratransit trips are customized services for people with health disabilities, who are unable to use regular transit services. Paratransit is expensive to operate because it requires advance reservations, which reduces travel flexibility. Data from a U.S. study show that, for paratransit users, aging has led to a 22% increase in the annual number of active users, but a 7% decrease in the annual number of trips taken (Garnier et al., 2022). Connecting paratransit to regular transit networks can alleviate the pressure on paratransit caused by the dramatic growth in elderly population. One study proposed a criterion for determining the priority of paratransit trips to regular transit network. The criteria include the paratransit user's autonomy, the universal design for stations and vehicles, the trip frequency, and the

walking distances from the origin and destination to the regular transit network (Garnier et al., 2022). This approach is a valuable exploration to the transfer between paratransit and regular transit, providing a solution to the deep aging in society. However, there are practical issues that need to be further addressed for implementing this standard. For example, the user's autonomy is not easily and accurately measured, and conventional transport must have a better inclusive design to enable the shift from paratransit.

5.2.3. Age-friendly informationalization

Older people's audiovisual, cognitive, and learning abilities are gradually declining. Existing studies have analyzed the navigation/wayfinding barriers and the smart technology gap in elderly mobility, and conducted age-friendly informationalization research in these two areas.

Wayfinding is a goal-directed process of information perception and decision-making, oriented to the identification of origin and destination routes. The process of way finding includes environmental perception, processing, decision-making, and route learning. Strategies for way finding include the simplest route, the shortest distance, pleasant surroundings, interesting features, avoidance of road intersections, less pollution and noise, and protection from bad weather (Bovy, 2009; Hou et al., 2021). Elderly wayfinding ability can be constrained by physical health status and transport environment (e.g., rush hour traffic, icy roads, road maintenance) (Wiener et al., 2009). During way finding, the elderly need to perceive and process spatial information to make route decisions. The key to age-friendly informationalization is easy to find, easy to be identified, and understandable.

In aspect of improving information comprehensibility, a study was conducted to analyze the factors that affect the disabled people's ability to navigate in walking environments, targeting different impairment types (e.g., visual, hearing, motor, cognitive, deafblind). This study proposed to help people with wayfinding impairments to get around by constructing cognitive maps, recognizing details of hazardous landmarks, and providing auditory signal instructions

(Prescott et al., 2020). In terms of improving information accessibility, one study suggested that centralized access to information should be provided, which means that the complete travel chain information can be accessed from a single point, including routes and schedules, accessibility levels, eligibility criteria, available assistance, travel times, etc. (Bjerkan and Øvstedal, 2018). Centralized access to information can reduce information search time and search difficulty for the elderly. For those with less severe audiovisual and comprehension impairments, a certain degree of information legibility and comprehensibility can be achieved through improved signage designing. One study proposed design criteria for inclusive transport guidance, timetables, contrasting colors, audio messages, and visual designing on large signs (Sze and Christensen, 2017). For the elderly with severe cognitive impairment, based on a socio-technical architecture for individuals with cognitive disabilities within a fixed-route public bus system, a study analyzed the implementation of the personal navigation for individuals with disabilities program (PNID). It found that the PNID program (i.e., AT service delivery process) combined with the WayFinder system (i.e., assistive technology) has the potential to meet the unique needs of individuals with cognitive disabilities when using PT (Harris et al., 2024). This case study provides an overview of an initial mixed approach to assisting older adults with cognitive disabilities in using the WayFinder system.

The revolution in transport information technology has reshaped individual accessibility, but the smart mobility paradigm has also brought inequitable social results (Zhang and Zhao, 2021). There is a digital divide in the use of smartphones among the elderly in terms of access, use, and knowledge. The digital divide reduces their outdoor activity frequency (Carney and Kandt, 2022) and exposes them to inequalities in access to telehealth services (Ramsetty and Adams, 2020). Infrastructure coverage and quality, smart device availability, and users' lack of knowledge and skills contribute to the digital divide among older people (Cottrill, 2018). For cell phone use, inequalities in the distribution of information and communications technology (ICT) lead to inequalities in the accessibility to transport information (Varghese and Jana, 2019). The above studies show that transport service digitization exacerbates the digital exclusion problems. These studies provide a new perspective on the critical understanding of the rising smart city/mobility paradigm (Kong and Woods, 2018; Vanolo, 2014). Bridging elderly digital divide is a critical step towards transport equity and is increasingly important to avoid their social isolation (Baker et al., 2018). Therefore, based on elderly knowledge structures and life experiences, retaining some traditional mobility services as a transition is necessary, while inclusive digital programs should be proposed to improve their accessibility to ICT.

Several studies have explored a variety of inclusive digital transport options. An inclusive smart bus stop is a public access point to a digital transport system that allows access to people who do not have an app or even a smartphone. The smart bus stop is considered an interconnected public space or IP-space. It can act as a travel assistant for older adults, providing them with centralized access to information,

helping them to understand real-time transit information and spatial congestion status, providing multi-modal or customized travel plans, rich news and information, and, especially, interconnected leisure activities by multilingual (Nápoles et al., 2020). The smart bus stop not only provides centralized access to transport information, but also serves as a public space vehicle that increases informational, emotional, and social connectivity for the elderly. Some cities are starting to develop smart bus stops or smart furniture to help older people integrate into digital transport systems (Table 10). A study analyzed a digital service program called DigiBy in Sweden Norrbotten County. It found that digital service solutions (DiSS) can reduce transport poverty among the elderly and create more flexible environments by increasing the likelihood that services will be available at a reasonable cost and in a reasonable time (Stjernborg and Svensson, 2024). Mobility as a Service (MaaS) can optimize travel by aggregating mobility information in an access-based system (Dadashzadeh et al., 2022). According to TRSE characteristics (Luz and Portugal, 2021) and MaaS prerogatives (Jittrapirom et al., 2017), a study developed an analytical framework for the relationship between TRSE and MaaS (Turbay and Firmino, 2023). This study provides a visualization pathway for understanding how MaaS mitigates TRSE. One study developed multidimensional out-of-home (OOH) activity clusters for six groups of older adults with different vulnerability types (Carney and Kandt, 2022), which could provide a more targeted contribution to the innovative services assessment for digital transport.

5.2.4. Cost affordability

PT cost has been identified as one of the factors contributing to transport exclusion and transport disadvantage, affecting elderly ability to access essential goods and services and maintain social connections (Lucas, 2012; Macintyre et al., 2008). Older people's sensitivity to costs has led to different fare policies significantly affecting their travel. For example, when the fare system changes from an area-based system to a distance-based system, the proportion of elderly riders decreases (Zhou et al., 2019). PT travel costs affect mobility, which can lead to social exclusion or inclusion (Stanley and Vella-Brodrick, 2009).

The main goal of introducing inclusive fare policies for the elderly is to address social exclusion. Around 100 places in the world offer free PT for older people, while others offer fares financed (Jones et al., 2013; Kębiowski, 2017), i.e., free bus passes (FBP) or concessionary bus passes (CBP). Eligibility for inclusive fares for the elderly is restricted by age, time slot or domicile (English National Concessionary Travel Scheme, 2017; Jackson et al., 2019; Mah and Mitra, 2017; UK Parliament, 2020; Zhang et al., 2019). Through these inclusive transport policies, governments hope to maintain elderly mobility and access to activities and services, which are related to their health and well-being (English National Concessionary Travel Scheme, 2017).

Existing studies have analyzed the effects of inclusive elderly fare policies, which are implemented in the United Kingdom, Australia, Canada, and China. As research has shown, free or subsidized PT as an active transport policy

Table 10 – Cases of centralized information access: smart bus stop/smart furniture (Nápoles et al., 2020).

Place	Smart bus stop/smart furniture
Paris, France	One stop, accessible to persons with disabilities, and providing free Wi-Fi and USB charge.
London, England	100 clear channel bus shelters, using Google Outside service to provide information.
Barcelona, Spain	Around 10 stops, with mobile-based payment system.
Budapest, Hungary	Ticket vending, parcel delivery, passenger counting, passenger information, wireless, USB charging, bike rental, air conditioning, taxi ordering, tourist information, news, advertisements, weather forecast, reverse vending, surveillance and other services.
Aizuwakamatsu, Japan	Multilingual displays will present live bus arrivals, timetables, route data, route transfers, service alterations (planned and unplanned) and a range of other travel advices.
Madrid, Spain	Automatic passenger count: increase the sensorization of bus to gain live data of bus occupancy, passenger flow and availability space for wheelchairs. Provide transport service: includes options for walking, cycling, private and public transport. Interconnected public space: a group of users can connect to people from other regions or countries in other IP-space nodes using the same or different languages. Everyone can participate remotely in different sports activities, physical activities, cultural activities and fun activities. Customized travel plan: customized trips for those who don't have a smartphone or don't have an Uber or taxi app and can pay with cash.

addresses the elderly inability to use PT due to financial pressures (Rye and Mykura, 2009). It is effective in promoting the PT use among the elderly (Reinhard et al., 2018). Transport subsidies provide many different cross-harbor options for the elderly in Australia. Free services have routes facilitating connections across the transport system (Rambaldini-Gooding et al., 2023). Streetcar fares in Hong Kong (China) are cheap and the elderly tend to give up their private cars in favor of PT travels (Szeto et al., 2017). In the United Kingdom, the FBP increased the PT use frequency among the elderly (Reinhard et al., 2019). Free buses in England increased “marginal trips” among older people under 75 years old, increasing their single-day trip frequency, travel distances, new destination types, and improvised travel destinations (Andrews et al., 2012). The elderly in the UK use buses more frequently for shopping due to the UK's inclusive fare policy (Mackett, 2014a). As a

Canadian study indicated, 38.2% of older adults felt that they would be impacted if the program discontinued. Particularly, those with lower incomes were 5.34 times more likely to be affected (Mah and Mitra, 2017). Australia's FBP encourages travel in the 70–74 age group, which are in the life stage of stopping driving (Andrews et al., 2012). This suggests that inclusive transport measures can convert elderly transport disadvantages into positive travel behaviors.

Inclusive fare policies have multiple externalities and positive social effects in terms of health promotion, social participation and well-being. First, inclusive fares significantly improve the physical and mental health for older adults. Physical activity is 43% higher among people with a bus pass than those without one (Webb et al., 2016). Inclusive fare policies can increase physical activity, reduce obesity (Laverty et al., 2018; Webb et al., 2012), improve cognitive function (Reinhard et al., 2019), reduce loneliness (Reinhard et al., 2018), and increase mental health (Andrews et al., 2012; Laverty et al., 2018; Reinhard et al., 2018), and decrease depressive symptoms (Jackson et al., 2019; Yang et al., 2020). The effectiveness of free bus policies on depressive symptoms varies by socioeconomic group, as well as by the type of residential segregation that varies by city income (greater in poorer centers than in wealthier centers) (Yang et al., 2020). Further, inclusive fare policies offer positive psychosocial dimensions. The elderly see free PT as symbolic for social equality, freedom and progress (Andrews et al., 2012). Second, inclusive fares promote social network maintenance and access to social capital for the elderly, reducing social segregation and social isolation. Free PT passes stimulate socialization among younger seniors (Andrews et al., 2012). This enables the elderly to maintain social networks, reduce social isolation (Barnes et al., 2016; Graham et al., 2020; Musselwhite, 2015; Shrestha et al., 2017), and mitigate social exclusion (Corran et al., 2018). People with licensed transit passes are about one-third less likely to be socially isolated (Jackson et al., 2019). Inclusive fare policies can increase elderly independence (Rambaldini-Gooding et al., 2023), allowing those who used to drive to maintain their autonomy when they reach the age banned from driving (Rambaldini-Gooding et al., 2023). By reducing costs, the elderly experience a significant association between PT access and social capital improvements (Eibich et al., 2016). Free bus travel makes them more able to engage in social activities and maintain healthy social relationships (Jackson et al., 2019). Third, inclusive fares can enhance elderly life quality, life satisfaction, and well-being. Free PT greatly improves elderly life quality (Andrews, 2011; Jackson et al., 2019), and the two are especially strongly associated among low-income populations (Rye and Mykura, 2009). CBP ownership and use are associated with higher life satisfaction (Jackson et al., 2019). They can significantly affect well-being by increasing physical activity and reducing social isolation (Jackson et al., 2019). The above literature review suggests that inclusive fares encourage PT travel among the elderly and have positive effects on their health, social participation, and well-being. It provides a cost-effective way to reduce the significant economic burden of morbidity associated with population aging and contributes to active aging. In this view, inclusive fares should be a key public health priority.

Free transit addresses the PT availability to some extent (Rambaldini-Gooding et al., 2021). However, the applicability and limitations of inclusive fares in different policy contexts must also be fully recognized, as free transit policies may be affected by other factors and fail to achieve the desired results. The literature review has found that factors affecting the effectiveness in inclusive fare policies are: rigid travel demand during non-free hours (Zhang et al., 2019), routes and schedules (Mah and Mitra, 2017), barriers to transport use (Rye and Carreno, 2008), negative attitudes towards PT (e.g., congestion) (Yang et al., 2020), health constraints (e.g., poor health and limited mobility), and substitution effect of structural changes in the transport system (Kandt and Leak, 2019). The policy effectiveness and its impact factors are in Table 11.

Therefore, the formulation of inclusive fare policies needs to comprehensively consider the impact by non-cost factors, so that the policy effect can be fully released in the practical application in cities. First, the inclusive fare policy should be formulated according to the travel characteristics of the elderly in the city where the policy is implemented. Second, PT capacity needs to be expanded. To ensure that low-income communities are included in the transit system, the relationship between free and subsidized PT and cross-transit systems, accessibility, and connectivity needs to be considered (Barrett et al., 2019). The FBP policy effects can be magnified by reducing bus wait times, increasing inclusive transit spaces, and improving attitudes toward buses among the elderly (Yang et al., 2020). Third, in addition to the commonly practiced FBP and CBP programs, more diverse fare systems and evaluation methods can be explored to coordinate stakeholders, so that fares can benefit more disadvantaged groups. A fair fare system must simultaneously consider the three criteria of benefits, costs and ability to pay (Nuworsoo et al., 2009). One study proposed a methodology and procedure for assessing fares and equity, with elements such as trajectory reconstruction, fare matching, segment labeling, visualization of required routes/stops, visualization of required routes/stops, commuter identification, and scenario analysis (Zhou et al.,

2019). These principles and programs provide a more detailed scheme for the fare policy development.

According to the literature review in Section 5, we conclude a framework for transport inclusion. Transport inclusion is a broad, overarching notion, which identifies a research/policy field and encompasses the following sub-concepts (Table 12).

6. Future research

The previous section has respectively reviewed the scholarship on elderly mobility research in five areas: theoretical foundations for transport inclusion, elderly mobility, the relationship between PT and active aging, transportation exclusion factors affecting mobility, and a research framework for transport inclusion. This section will further incorporate spatial production theory, elderly mobility, capability approach (CA), and subjective and objective accessibility studies to look forward to challenges and opportunities for the research on elderly PT inclusion in the future.

6.1. Examining the mechanisms for the impact by perceived transport inclusion on active aging

Existing research (Section 4.2) clarifies three aspects of the impact by transport inclusion on active aging, namely health, social participation, and well-being. However, how transport inclusion has systematically affected active aging is not clear. Identifying the mechanisms by which transport inclusion affects active aging is a key issue, when planning and designing PT to improve elderly inclusion. Existing research on the relationship between PT and active aging is generally based on the behavioral-social effects paradigm, and lacks in-depth research on attitudinal-social effects.

Existing research has the following limitations: first, although a small body of literature has focused on the relationship between travel satisfaction and well-being and discussed the impact by subjective transport evaluations on life domains, yet travel satisfaction and perceived transport inclusion have different connotations. It is because transport inclusion addresses the transport service attributes that are of concern to the disadvantaged traveler groups and whether these services are inclusive of their capabilities. In contrast, travel satisfaction focuses only on the service attributes of regular transport. Second, there are differences in how different groups perceive the same transport system. For example, the elderly have lower levels of perceived accessibility to PT than younger people (Lattman et al., 2016a, 2016b). Therefore, there is a need to focus on their perceptual differences separately in order to increase inclusiveness for different disadvantaged groups in terms of their travel needs and capabilities. Existing perception studies related to TRSE have found the effect by transport on positive emotions in commuter groups (Singleton, 2019) and the effect by perceived travel time on accessibility poverty (Dharmowijoyo et al., 2020), but the perception concept has not yet been emphasized in the context of research questions on the elderly. Third, perception is a concept closely related to social inclusion, which has not yet been focused on in the study on the relationship between transport and active

Table 11 – Policy effectiveness and its impact factors.

Policy	Detail
Effectiveness of the policy	Physical and mental health: increase in physical activity, less depression, better mental health. Social participation: maintaining social networks and social capital, reducing isolation and segregation, maintaining independence and autonomy. Well-being: improving life quality and satisfaction.
Impact factors of the policy	Individual factors: health constraints, travel during non-free hours, negative attitudes towards PT. Transport capacity factors: barriers to transport use, routes and schedules. Transport structure factors: tram extension, emerging ride-sharing and e-hailing services.

Table 12 – The framework for transport inclusion.

Framework	Notion and definition	Reference
Opportunity accessibility	• System accessibility: the distance from a neighborhood to a transit station. • System facilitated accessibility: travel time cost from neighborhoods to the destination. • Integral accessibility: the number of opportunities that can be reached by the transit system within a certain time or distance range. • Relative accessibility: the accessibility of PT trips in terms of travel time and cost compared to other travel modes.	Lei and Church (2010), Malekzadeh and Chung (2020), Wang and Chen (2015)
Physical accessibility	• Inclusive PT system: physical accessibility and safe of vehicles, stations, stops, platform, indications and support. • Inclusive street: accessibility, connectivity, streetscape, safety and security, urban vitality, and greenery. • Paratransit trips: customized services for people with health disabilities, who are unable to use regular transit services.	Garnier et al. (2022), Wang et al. (2019)
Age-friendly informationalization	• Easy to find: provide centralized access, ICT cover and inclusive smart bus stop. • Easy to be identified: apply barrier free design criteria for visual or hearing impaired person. • Understandable: provide personal navigation with service processes and assistive technology for cognitive impaired person.	Bjerkan and Øvstedal (2018), Harris et al. (2024), Napoles et al. (2020), Sze and Christensen (2017)
Cost affordability	• Inclusive price: free bus passes or concessionary bus passes with less limitation. • Affordable price: travel costs is affordable to obtain the geographic range of social opportunity.	Church et al. (2000)

aging for the elderly. In the conceptual system of social inclusion, perception is one of the main ways to evaluate social inclusion. As a social public space, the elderly's perception of social trust, acceptance and recognition by others, community group gap, and social fairness in transport space will affect their subjective sense of well-being. Existing research has focused on the relationship between perceived psychosocial factors and social inclusion, but the mechanisms by which perceptions on transport inclusion affect active aging are unclear.

By emphasizing the role of transport space on social perceptions, spatial production theory provides a theoretical basis for examining the mechanisms by which perceptions on elderly PT inclusion affect active aging. Accompanying with the social turning of traditional geospatial studies, space has been recognized as participating in the production of social culture and ideology. From the spatial perception perspective, Lefebvre's spatial production theory explains how people's perception of the spatial environment affects their perception of the social life sphere, explaining the mechanism by which transport inclusion/exclusion produces social inclusion/exclusion. Spatial practice is the perception on environmental space, spatial reproduction is the conceptualization about social space, and reproduced space is the mapping out of real life (Wang, 2009). Transport is not only a tool for passengers' spatial displacement, but also a mediator for passengers' perception on social inclusion in their travels. Social members perceive the value orientation of transport concepts (i.e., spatial reproduction) from their travels (i.e., spatial practice) and form judgments about their social environment (i.e., reproduced space). Transport spatial reproduction is the presentation of a certain transport concept fixed in the transport environment through planning layout, transport service and environmental design. Transport spatial practice is that people assess

whether the transport environment is friendly or not, inclusive or exclusive to them through their travel feeling in the transport space (e.g., opportunity accessibility, environment friendliness, image reputation, etc.). The reproduced space in transport domain is a presentation of the transport-related social effects. This means that the spatial value of transport does not only remain in the role of the transport environment as a safeguard for aging society, but can also construct a higher meaning about well-being. Transport inclusion is constructed in the transport space and transmitted to the social life sphere through the value perception on the travel process of the disadvantaged groups, thus affecting their life satisfaction and well-being.

In addition, a number of existing transport perception studies provide insights into variables and measures for a deeper understanding about elderly PT inclusion. First, existing research findings support the impact by transport perceptions on travel behaviors and attitudes. People decide whether to use transport services based on the expected availability, accessibility, affordability, and attractiveness of the service (De Paepe et al., 2023). Existing research has found that the perceived pension value for the elderly is a strong driver against road pricing (Nikitas et al., 2018). Older people's perception about PT safety significantly impacts their PT trips (Morris et al., 2017). Perceived travel time in PT trips directly affects passenger satisfaction (Tiznado-Aitken et al., 2021). An evaluation on PT services for older passengers discussed passenger expectation, perceived quality, and perceived value, relationship between passenger satisfaction, passenger complaint, and passenger loyalty (Yuan et al., 2019). Second, some of the perceived variables of community environment are instructive for future research on transport inclusion perceptions. These studies have identified perceived factors affecting the satisfaction with community environments, which include:

perceptions on accessibility, perceived safety and fear of crime, perceptions on public space quality, perceived esthetic quality, and perceived quietness (Mouratidis, 2021). These variables also provide new insights into the evaluation on transport inclusion perceptions. Third, a number of psychosocial variables are instructive for analyzing the relationship between transport inclusion and society. Perceived social norms, social support, and neighborhood social context are fundamental drivers of social inclusion, and addressing these psychosocial factors can improve older women's perceptions on PT accessibility and social inclusion (Al-Rashid et al., 2021). Social norms for PT use affect social exclusion/inclusion among the elderly, with perceived accessibility moderating the impact by social and personal norms on social exclusion/inclusion (Al-Rashid et al., 2023b). Perceived accessibility, social norms, attitudes, and social support are significantly negatively related to social exclusion (Al-Rashid et al., 2023a). Fourth, some studies have developed perceived accessibility scales (PAC) to capture riders' evaluations on PT (Lattman et al., 2016a, 2016b). Although the above research literature has explored perceptual evaluations, the research question is only at the level of transport service evaluation and lacks an extension to the connotation of social inclusion or active aging. Transport inclusion is a multidimensional concept, and future research needs to be deepened in multiple dimensions of transport inclusion, such as perceived accessibility, perceived spatial accessibility, perceived age-friendly informationalization, and perceived cost affordability.

6.2. Expanding the multilevel research space for CA

This study reviewed the theoretical foundations for transport inclusion (Section 3). As a transport evaluation principle, transport equity theory bridges the gap between utility theory and works to address the mobility dilemma faced by the disadvantaged traveler groups. John Rawls' justice theory and Walzer's spheres of justice provide the philosophical foundation and distribution principles for transport inclusion. However, how to apply these ethical principles to the practice for transport inclusion is still a challenge; John Rawls' maximum principle guarantees fairness in transport resource distribution by setting maximum and minimum constraints and maximizing the number of accesses. This approach is suitable for considering the equity of transport resource distribution at the macro level, but it cannot answer the question of the impact by individual capability and transport resource possession on the social opportunity differences at the micro level. Because of the neglect on the capability differences, this approach is also not conducive to the precise policy and accurate assessment for different disadvantaged traveler groups in practice. Therefore, how elderly individual abilities contribute to the inequality and how to accurately assess their differences in life chances based on individual ability as well as resource differences still need to be explored in depth.

The capability approach, proposed by Amartya Sen, has provided theoretical support for addressing micro-individual assessments. It argues that social equity should be evaluated

in terms of human capabilities, not just basic social goods (Sen, 1992). CA treats capabilities as individual rather than social responsibilities, making it more difficult to recognize other direct and indirect impacts on health, environment, and well-being from transport systems (Randal et al., 2020). In contrast to other studies, CA conceptualizes transport (policy) as a conversion factor, meaning that the general public can be facilitated to convert capabilities into opportunities, which include employment, education, health, and participation (Randal et al., 2020). CA understands an individual freedom level in terms of five characteristics: resources, conversion factors, capabilities, choices, and functioning (Table 13). Conversion factors determine the extent to which a person actually utilizes the available resources. Conversion factors may be multiple and correlating. For example, if an older person has limited mobility and also has a low income, his conversion factor for using transport to access resources will be doubly affected.

CA offers new ways of thinking about assessing transport equity and has a number of unique advantages. First, based on it, transport policy can be conceptualized as a social conversion factor. Multiple conversion factors shape a person's capability to dominate transport and social resources, which is a natural selection mechanism that determines which opportunities are available to which people (Vecchio and Martens, 2021). Based on the travel capabilities, extracting the characteristics in disadvantaged traveler groups facilitates quantitative research at a more micro level and allows for an accurate assessment on their capabilities to convert resources under the impact by correlating factors for a given individual, rather than for a particular group of individuals. Second, the utility theory, which is utilitarian value oriented, advocates the principle of maximizing the welfare of transport benefits or overall social interests to set policy priorities. In contrast, under a humanistic value-oriented principle, CA advocates that mechanisms for evaluating transport in terms of individual capabilities should be established, arguing that interventions are necessary if individuals fall below critical advantage thresholds, even if these interventions imply redistribution

Table 13 – Elements of the capability approach (Vecchio and Martens, 2021).

Element	Detail
Resource	Resources are a means to achievement.
Conversion factors	Conversion factors are a person's capability to convert resources into social opportunities.
Capabilities	The capabilities are the result of a combination of resources and conversion factors. The larger the capability set the greater the choice range available to the individual.
Choices	Choices are the opportunities and resources available to the individual.
Functionings	Functionings are the states of being (i.e., achieving “to be” or “to do”) that an individual can ultimately attain.

on benefits and losses in a utilitarian sense (Thomopoulos, 2013). Third, in contrast to Rawls (1971) maximization criterion, which is a holistic assessment of the highest and lowest constraint ranges, averages, and minimums, Martens proposes a similar line of thought. He suggests that equitable transport investment projects need to generate disproportionate benefits for low mobility groups, so that the gap between areas or communities with the lowest and highest levels of accessibility should be kept within predefined limits (spatial correlation or inter-community equity) (Martens et al., 2012). The maximization criterion quantifies the ethical principles of transport equity and refines the understanding about the basic rights of advantaged and disadvantaged people at the practical level, but it cannot specifically characterize different disadvantaged traveler groups. The capability approach, on the other hand, can break down different disadvantaged traveler groups into specific capability elements and can quantify the correlating capability factors for disadvantaged traveler group, increasing the granularity for transport inclusion evaluation.

Knowledge findings from health sciences and transport sciences identified elderly travel capability characteristics. These characteristics include: health dimensions (physical adaptability and psychological adaptability), information capability dimensions (path finding capability and smartphone usage capability), transport usage capability dimension (vehicle usage capability). On the health dimension, speeding vehicles and complex environments increase older people's feelings of their vulnerability or fear of traveling (De Donder et al., 2013). A reduced sense of control over the environment inhibits their willingness to act (Downey and Van Willigen, 2005). The elderly who lack experience in long-distance travel show negative self-evaluation and anxiety before traveling, and older women are more likely to have a fear of getting lost (Gayman et al., 2008). Due to the slowness in movement, they feel the invisible pressure of waiting from other passengers. Weaker herd immunity increases their travel anxiety (Liu et al., 2021b). In terms of the information dimension, the complexity of connecting urban PT routes and the spatial design in subway stations cause problems for the elderly in terms of route cognition. Despite the many advantages of smartphone travel services, it is difficult for the elderly to benefit from technological innovations, and they have a sense of powerlessness towards digital transport services (Liu et al., 2021a, 2021b). In terms of transport usage capability, mastery of multiple transport modes, especially long-distance transport, is a constraint affecting the radius of their social activities. There is a clear link between holding a driver's license, access to PT and ability to travel (Ryan et al., 2019). Older people's ability to learn about different transport modes has important implications for their travel characteristics (He et al., 2009). In particular, new transport apps and technologies are rapidly emerging, and the impact from new technologies on elderly mobility is unclear (Zhang and Zhao, 2021). While existing studies have identified the existence of different mobility disadvantages for the elderly, researchers have not yet reached a consensus on how to quantitatively study CA on issues related to transport and its social impacts.

Some scholars have tried to make new explorations on CA studies, which provide new directions for thinking about the CA application in the field of elderly PT inclusion. Based on the basic CA elements, a CA study was quantified using dynamic Bayesian networks. The study identified the individuals at high risk of social exclusion by comparing their capabilities. This is one of the more comprehensive quantitative studies to employ the CA framework, modeling in a probabilistic manner the different components that shape an individual's ability to access opportunities (Bantis and Haworth, 2020). However, this CA study has troubles in the under representation of demographic groups at high TRSE risk in the sample, and the inability to capture the actual needs behind an individual's travel choices. One study has proposed a dual assessment approach to transport equity based on conversion factors, including a top-down approach and a bottom-up approach. Top-down approach: accessibility-based analysis of transport and land use systems. Bottom-up approach: assessments of a person's perceived accessibility to opportunities (Vecchio and Martens, 2021). The former has the advantage of assessing the number of opportunities, but will not provide insights into non-quantitative issues (e.g., fares, safety, etc.). The latter, on the other hand, can develop more complete explanations in transport equity, e.g., by examining how capabilities are formed or used, or by examining transport and social inclusion in two parallel dimensions, or by complementing some of the less easily detected conversion factors of capabilities. One study developed a capability index (CI) using fuzzy AHP techniques to assess the prioritization for urban projects, and showed that individuals with higher capabilities benefit from a better community environment, with the average CI for residents of central business districts being 20% higher than that of suburban residents (Azmoodeh et al., 2022). By defining five elderly travel capabilities, one study found that travel capability was a moderator for the perceived transport inclusion value and for health, social participation, and subjective well-being (Zhang et al., 2024). In accessibility research, a large number of research results focus on location-based accessibility as well as accessibility based on spatio-temporal constraints, and there is still room for research on accessibility based on individual mobility capabilities. Based on CA theory, one study stated that, for equity analysis, accessibility should be defined as the ease of personal travel rather than the aggregation of proximate activities (Azmoodeh et al., 2022). Through variables such as accessibility, transit availability, ability to use transit, and ideal travel choice versus actual travel, an individual's ability (i.e., freedom or opportunity) and functioning (i.e., achievement) can be assessed.

Based on CA, assessing the capabilities and opportunities for disadvantaged traveler groups is a valuable direction to take. The questions of how mobility capabilities affect the actual access of disadvantaged groups to social opportunities and how increased social opportunities affect social inclusion remain unresolved (Zhang and Zhao, 2021). In particular, the elderly population is highly heterogeneous, and CA can provide more accurate feedback on their multifactorial transport exclusion, examine how conversion factors and

conversion mechanisms play their roles, and examine how mobility affects transport inclusion and active aging in an integrated manner. These issues still need to be continuously explored in the future.

6.3. Exploring an integrated subjective-objective research model for elderly PT inclusion

The recent rise of big data-driven accessibility studies tends to exclude the elderly from the data, which poses a challenge to accurately measure their mobility needs. Cell phone signaling and smart card datasets can provide an accurate and immediate measure of people's mobility, which can help identify the spatial and temporal characteristics of mobility for certain social groups. However, the elderly are a vulnerable group using digital technology and the disadvantages of big data approaches are obvious. One study analyzed the potential impact from the rise of data-driven decision-making on elderly mobility. The study showed that elderly mobility is an area where there are significant gaps in available data (Sourbati and Behrendt, 2020). This research suggests that policy makers should develop pluralistic approaches to data collection and analysis in order to address digital exclusion in smart public services.

Compared to a single data source, a combination of subjective and objective data sources provides a more comprehensive understanding for studying the mechanisms by which transport inclusion affects society. As people have different adaptive capabilities to different environments, objective and subjective indicators often show differences (Cummins, 2000). Such differences may be caused by an individual's economic characteristics, demographic attributes, and travel capabilities (Lattman et al., 2016a, 2016b). A study combined objective data (transport data) with subjective data (time satisfaction) in an accessibility study, and compared time satisfaction in four travel modes as an impedance coefficient. Based on a gravity standardized measure, it revealed the differences between the accessibility indicators, with and without considering the satisfaction (Chaloux et al., 2019). This finding emphasizes the importance of incorporating subjective evaluations into accessibility models. One study found that both subjective and objective environments shaped community satisfaction, which in turn contributed to subjective well-being. But perceived environmental characteristics were more strongly correlated with community satisfaction (Lee et al., 2017). Using a combination of subjective and objective data sources, a study examined the impact by actual travel behavior and potential travel (i.e., perceived level of daily travel) on travel satisfaction and life quality among the elderly at different ages. Its results identified the importance of mobility on life quality, from both objective and subjective perspectives (Kim et al., 2020).

In addition, quantitative and qualitative research each has its own strengths. Existing research has under-emphasized qualitative research methods. For highly heterogeneous groups, qualitative research methods have unique value. In the future, there is a need to continue to exploring the potential of qualitative research methods to gain a deeper understanding into elderly needs. In general, quantitative

research uses hypothesis testing to explore conclusions with generalizable value, but it is easy to ignore differences in geography, demographic characteristics, social norms, and living habits. However, these factors have more comprehensive explanatory power and policy contextual value in revealing the relationship between transport and active aging. The open and exploratory nature of qualitative research helps to identify new research variables. For example, traditional modeling of transport accessibility usually aims at the shortest travel time or distance, ignoring the details on travel characteristics for different groups. In contrast, qualitative research helps to explore behavioral details which may constitute new accessibility travel impedance factors or accessibility research questions for the elderly. Qualitative analysis has the potential to build bridges to quantitative research (Marx et al., 2014), yielding new findings on a larger scale.

7. Conclusions

Transport is a key factor in achieving social inclusion (Currie and Stanley, 2008). Transport inclusion is a key indicator for evaluating age-friendly cities (Gawlak et al., 2021; Ivan et al., 2024). Transport inclusion policies have many positive externalities (Swift et al., 2021).

Over the past 30 years, research on transport and its social impacts has experienced rapid growth. For countries already aging or about to age, focusing on PT inclusion in aging societies will not only benefit the elderly, but also many types of disadvantaged traveler groups, and is a fundamental safeguard for society to remain vibrant.

This study provides an overview of the major areas of research on elderly mobility in recent years. Based on an extensive literature overview, our study reviews the multidisciplinary theoretical underpinnings of transport inclusion, as well as four key empirical research themes, including elderly characteristics and mobility, research on the relationship between transport and active aging, the identification of elderly transport exclusion, and a framework for evaluating inclusive PT transport policies. Through a literature review, this study analyzes valuable findings in the field of elderly mobility research and proposes valuable future research directions. This paper contributes to a systematic understanding towards the mobility challenges faced by aging societies and the indirect impact from active transport inclusion policies on aging societies.

This study has showed that the elderly are a highly heterogeneous and vulnerable group, experiencing multiple barriers to mobility. Mobility affects their aging status, including health, social participation, and subjective well-being. Transport factors contributing to elderly social exclusion are accessibility exclusion, physical exclusion, information exclusion, and economic exclusion. The policy framework for transport inclusion should be established from four aspects: opportunity accessibility, physical accessibility, age-friendly informationalization, and cost affordability. Future research should pay attention to examining the mechanism of the impact by perceived transport inclusion on active aging, expanding the multilevel CA research space, and

exploring the subjective-objective integrated PT inclusion research model.

However, this study has some limitations. The evaluation framework for transport inclusion proposed in this paper is based on the prevalent PT mobility constraints among the elderly. This is a good reference value for the elderly and cities with predominantly PT travel modes, but for a small number of cities or villages with dispersed residences or high driving rates among the elderly, the development of transport inclusion policies for other travel modes may be a more feasible approach. It should be noted that, given the multifactorial correlation of elderly transport disadvantages, this paper includes individually informative research literature on other disadvantaged traveler groups in the analysis. This approach to selecting literature relaxes the scale in terms of conceptual accuracy, but from the perspective of transport inclusion policy implementation, it has a more holistic consideration contributing to the development of a transport system inclusive of all.

Based on the above research, this paper makes three recommendations for transport policy makers to address the challenges of aging societies.

First, the elderly are a group with high heterogeneity, and transport inclusion requires a multidimensional policy framework to construct a comprehensive and adequate age-friendly transport system. It is important to optimize the distribution of transport and opportunity resources for older populations, enhance the prevalence of accessible facilities, and examine the effectiveness in implementing free or subsidized PT policies. In particular, in the context of fully digitalized mobility in cities, the establishment of a mobility system that takes into account the traditional travel habits among older users or an age-friendly digital service access is an urgent need to be addressed. This will prevent the elderly from being excluded from the digital society.

Secondly, in order to better utilize the positive effects of the built transport environment in the social sphere, the evaluation on elderly perception on the transport environment should not be ignored. This helps policy makers to accurately identify the travel behavior characteristics among the elderly and provide targeted transport services to eliminate transport disadvantages. In addition, self-reported data from the elderly and objective measurements are equally important, and comprehensive subjective and objective models provide more comprehensive feedback on the effects of transport inclusion policies.

Third, active aging encourages the elderly to return to society in a comprehensive manner, and macro-level transport inclusion policymaking needs to focus on the structural changes in the opportunity needs of older people in society as a whole. These needs include not only living needs, but also productive needs and personal development needs. These elderly opportunity needs are influenced by local concepts of old age or socio-demographic structure, and may be characterized differently in different countries or regions. Compared with western societies, Chinese older people view childcare more as an obligation and adopt a more home-based model of aging. If a society enters a heavy aging stage, it means that the socio-demographic structure is imbalanced and there is a scarcity in labor force, and the return to the workplace for the

elderly will be a social demand. Therefore, policy decisions on transport inclusion should be adjusted accordingly to the structural changes in the opportunity needs of the elderly.

Conflict of interest

The authors do not have any conflict of interest with other entities or researchers.

Acknowledgments

The research was supported by the Key Projects of Central Universities: Research on the Theory and Practice of Transport Inclusion for Active aging (No. 300102112601); the Transformation Program of Scientific and Technological Achievements of Sichuan Province: Intelligent Scheduling System of Dynamic Logistics Based on Complex Network (No. 2021ZHCG0008).

REFERENCES

- Accedes-Gonzalez, C., Cook, S., May, A., 2015. Bus use in a developing world city: implications for the health and well-being of older passengers. *Journal of Transport & Health* 2 (2), 308–316.
- Achuthan, K., Titheridge, H., Mackett, R.L., 2010. Mapping accessibility differences for the whole journey and for socially excluded groups of people. *Journal of Maps* 6 (1), 220–229.
- Adorno, G., Fields, N., Cronley, C., et al., 2018. Ageing in a lowdensity urban city: transportation mobility as a social equity issue. *Ageing & Society* 38 (2), 296–320.
- Agost-Felip, R., Ruá, M.J., Kouidmi, F., 2021. An inclusive model for assessing age-friendly urban environments in vulnerable areas. *Sustainability* 13 (15), 8352.
- Allen, J., Farber, S., 2019. Sizing up transport poverty: a national scale accounting of low-income households suffering from inaccessibility in Canada, and what to do about it. *Transport Policy* 74, 214–223.
- Allen, J., Farber, S., 2020. Planning transport for social inclusion: an accessibility-activity participation approach. *Transportation Research Part D: Transport and Environment* 78 (3), 102212.
- Al-Rashid, M.A., Goh, H.C., Harumain, Y.A.S., et al., 2020. Psychosocial barriers of public transport use and social exclusion among older adults: empirical evidence from Lahore, Pakistan. *International Journal of Environmental Research and Public Health* 18 (1), 185.
- Al-Rashid, M.A., Harumain, Y.A., Goh, H.C., et al., 2021. Psychosocial factors of public transport users and social inclusion implications among older women in Pakistan. *Journal of Urban Planning and Development* 147 (4), 04021046.
- Al-Rashid, M.A., Harumain, Y.A.S., Goh, H.C., et al., 2023a. Psychosocial barriers to public transport use triggers older adults' social exclusion: empirical evidence from Pakistan. *Journal of Urban Planning and Development* 149 (1), 04022050.
- Al-Rashid, M.A., Harumain, Y.A.S., Goh, H.C., et al., 2023b. Perceived norms of public transport use as the determinants of older adults' social exclusion: evidence from Pakistan. *Cities* 137, 104264.

- Andrews, G., 2011. Just the Ticket? Exploring the Contribution of Free Bus Fares Policy to Quality of Later Life (PhD thesis). University of the West of England, Bristol.
- Andrews, G., Parkhurst, G., Shaw, J., 2012. The grey escape: investigating older people's use of the free bus pass. *Transportation Planning and Technology* 35 (1), 3–15.
- Ang, B.H., Oxley, J.A., Chen, W.S., et al., 2019. To reduce or to cease: a systematic review and meta-analysis of quantitative studies on self-regulation of driving. *Journal of Safety Research* 70, 243–251.
- Asadi-Shekari, Z., Moeinaddini, M., Shah, M.Z., 2013. Disabled pedestrian level of service method for evaluating and promoting inclusive walking facilities on urban streets. *Journal of Transportation Engineering* 139 (2), 181–192.
- Australian Social Inclusion Board, 2012. *Social Inclusion in Australia: How Is Australia Fairing?* second ed. Australian Government Department of Families, Housing, Community Services and Indigenous Affairs, Canberra.
- Azmoodeh, M., Haghghi, F., Motieyan, H., 2022. Capability index: applying a fuzzy-based decision-making method to evaluate social inclusion in urban areas using the capability approach. *Social Indicators Research* 165 (8), 77–105.
- Baker, S., Warburton, J., Waycott, J., et al., 2018. Combatting social isolation and increasing social participation of older adults through the use of technology: a systematic review of existing evidence. *Australasian Journal on Ageing* 37 (3), 184–193.
- Bantis, T., Haworth, J., 2020. Assessing transport related social exclusion using a capabilities approach to accessibility framework: a dynamic Bayesian network approach. *Journal of Transport Geography* 84, 102673.
- Barnes, J., Morris, A., Welsh, R., et al., 2016. Injuries to older users of buses in the UK. *Public Transport* 8 (1), 25–38.
- Barrett, S., Gariban, S., Belcher, E., 2019. *Fair Access: Towards a Transport System for Everyone*. Centre for London, London.
- Batty, M., 1976. *Urban Modelling: Algorithms, Calibrations, Predictions*. Cambridge University Press, Cambridge.
- Bergstad, C.J., Gamble, A., Gärling, T., et al., 2011. Subjective well-being related to satisfaction with daily travel. *Transportation* 38 (1), 1–15.
- Bertolini, L., Le Clercq, F., Kapoen, L., 2005. Sustainable accessibility: a conceptual framework to integrate transport and land use plan-making. Two test-applications in The Netherlands and a reflection on the way forward. *Transport Policy* 12 (3), 207–220.
- Bishop, N.J., Eggum-Wilkens, N.D., Haas, S.A., et al., 2016. Estimating the co-development of cognitive decline and physical mobility limitations in older U.S. adults. *Demography* 53 (2), 337–364.
- Bjerkkan, K.Y., Øvstedal, L.R., 2018. Functional requirements for inclusive transport. *Transportation* 47, 1177–1198.
- Bocarejo, S.J.P., Oviedo, H.D.R., 2012. Transport accessibility and social inequities: a tool for identification of mobility needs and evaluation of transport investments. *Journal of Transport Geography* 24, 142–154.
- Bovy, P.H.L., 2009. On modelling route choice sets in transportation networks: a synthesis. *Transport Reviews* 29 (1), 43–68.
- Brandt, L., Liu, S., Heim, C., et al., 2022. The effects of social isolation stress and discrimination on mental health. *Translational Psychiatry* 12 (1), 398.
- Brandtzæg, P.B., Heim, J., Karahasanović, A., 2011. Understanding the new digital divide—a typology of internet users in Europe. *International Journal of Human-Computer Studies* 69 (3), 123–138.
- Button, K., Nijkamp, P., 1997. Social change and sustainable transport. *Journal of Transport Geography* 5 (3), 215–218.
- Buys, L., Snow, S., Van Megen, K., et al., 2012. Transportation behaviours of older adults: an investigation into car dependency in urban Australia: transportation behaviours of older adults. *Australasian Journal on Ageing* 31 (3), 181–186.
- Cantekin, Ö.F., Arpacı, F., 2023. The relationship between social inclusion and life satisfaction among Turkish older people. *Educational Gerontology* 50 (1), 1–10.
- Carney, F., Kandt, J., 2022. Health, out-of-home activities and digital inclusion in later life: implications for emerging mobility services. *Journal of Transport & Health* 24, 101311.
- Casal, P., 2007. Why sufficiency is not enough. *Ethics* 117 (2), 296–326.
- Cassarino, M., Setti, A., 2016. Complexity as key to designing cognitive-friendly environments for older people. *Frontiers in Psychology* 7, 1329.
- Chaloux, N., Boisjoly, G., Grisé, E., et al., 2019. I only get some satisfaction: introducing satisfaction into measures of accessibility. *Transportation Research Part F: Traffic Psychology and Behaviour* 62, 833–843.
- Chen, Y., Bouferguene, A., Al-Hussein, M., 2018. Neighborhood design and regional accessibility of age-restricted communities from resiliency and spatial perspectives. In: *Construction Research Congress*, New Orleans, 2018.
- Chen, Y., Liu, B., Shen, Y., et al., 2022. Assessing accessibility to service facilities for older people in age-restricted communities from the perspective of equity. *Journal of Transport & Health* 27, 101515.
- Chen, Z., Yeh, A.G.O., 2019. Accessibility inequality and income disparity in urban China: a case study of Guangzhou. *Annals of the American Association of Geographers* 109 (1), 121–141.
- Cheng, L., Caset, F., Vos, J.D., et al., 2019. Investigating walking accessibility to recreational amenities for elderly people in Nanjing, China. *Transportation Research Part D: Transport and Environment* 76, 85–99.
- Cheng, S., White, M., Abraham, C., et al., 2016. Commuting and wellbeing in London: the roles of commute mode and local public transport connectivity. *Preventive Medicine* 88, 182–188.
- Cheng, L., Yang, M., Vos, J.D., et al., 2020. Examining geographical accessibility to multi-tier hospital care services for the elderly: a focus on spatial equity. *Journal of Transport & Health* 19, 100926.
- Church, A., Frost, M., Sullivan, K., 2000. Transport and social exclusion in London. *Transport Policy* 7 (3), 195–205.
- Corran, P., Steinbach, R., Saunders, L., et al., 2018. Age, disability and everyday mobility in London: an analysis of the correlates of “non-travel” in travel diary data. *Journal of Transport & Health* 8, 129–136.
- Cottrill, C.D., 2018. *Data and Digital Systems for UK Transport: Change and Its Implications*. Available at: <https://www.gov.uk/government/publications/future-of-mobility-data-and-digital-systems-for-uk-transport> (Accessed 1 February 2024).
- Council of the European Union, 2004. *Employment, Social Policy, Health and Consumer Affairs*. EU, Brussels.
- Crotti, D., Maggi, E., Pantelaki, E., et al., 2021. Public transport use and health status in later life: which relationship? *Research in Transportation Business and Management* 40, 100591.
- Cummins, R.A., 2000. Objective and subjective quality of life: an interactive model. *Social Indicators Research* 52 (1), 55–72.
- Currie, G., Delbosc, A., 2010. Modelling the social and psychological impacts of transport disadvantage. *Transportation* 37, 953–966.
- Currie, G., Richardson, T., Smyth, P., et al., 2010. Investigating links between transport disadvantage, social exclusion and well-being in Melbourne—updated results. *Research in Transportation Economics* 29 (1), 287–295.
- Currie, G., Stanley, J., 2008. Investigating links between social capital and public transport. *Transport Reviews* 28 (4), 529–547.
- Dai, D., 2011. Racial/ethnic and socioeconomic disparities in urban green space accessibility: where to intervene? *Landscape and Urban Planning* 102 (4), 234–244.

- Dai, B., Ware, W.B., Giuliani, C.A., 2012. A structural equation model relating physical function, pain, impaired mobility (IM), and falls in older adults. *Archives of Gerontology and Geriatrics* 55 (3), 645–652.
- Dadashzadeh, N., Woods, L., Ouelhadj, D., et al., 2022. Mobility as a service inclusion index (MaaSINI): evaluation of inclusivity in MaaS systems and policy recommendations. *Transport Policy* 127, 191–202.
- Davidson, P., 2008. Land Use and Public Transportation Accessibility Index (LUPTAI) Model Improvements. Queensland Transport Integrated Transport Planning Division, Brisbane.
- Dávila, M.C., Díaz-Morales, J.F., 2009. Age and motives for volunteering further evidence. *Europe's Journal of Psychology* 5 (2), 82–95.
- Deboosere, R., El-Geneidy, A., 2018. Evaluating equity and accessibility to jobs by public transport across Canada. *Journal of Transport Geography* 73, 54–63.
- De Donder, L., Buffel, T., Dury, S., et al., 2013. Perceptual quality of neighbourhood design and feelings of unsafety. *Ageing and Society* 33, 917–937.
- Delbosc, A., Currie, G., 2011. Transport problems that matter: social and psychological links to transport disadvantage. *Journal of Transport Geography* 19 (1), 170–178.
- Demnitz, N., Hogan, D.B., Dawes, H., et al., 2018. Cognition and mobility show a global association in middle- and late-adulthood: analyses from the Canadian longitudinal study on aging. *Gait & Posture* 64, 238–243.
- De Paepe, L., Van Acker, V., Witlox, F., 2023. To share or not to share, by whom is the question. Acceptability and acceptance of shared transport services by vulnerable groups. *Transport Reviews* 43 (5), 935–969.
- Department for Communities and Local Government, 2008. The English Indices of Deprivation 2007. Department for Communities and Local Government (DCLG), London.
- De Vos, J., Schwanen, T., Van Acker, V., et al., 2013. Travel and subjective well-being: a focus on findings, methods and future research needs. *Transport Reviews* 33 (4), 421–442.
- Dharmowijoyo, D.B.E., Susilo, Y.O., Syabri, I., 2020. Time use and spatial influence on transport-related social exclusion, and mental and social health. *Travel Behaviour and Society* 21, 24–36.
- Di Gessa, G., Grundy, E., 2014. The relationship between active Ageing and health using longitudinal data from Denmark, France, Italy and England. *Journal of Epidemiology and Community Health* 68 (3), 261–267.
- Doi, K., Sunagawa, T., Inoi, H., et al., 2016. Transitioning to safer streets through an integrated and inclusive design. *IATSS Research* 39 (2), 87–94.
- Dominy, N., Kempson, E., 2006. Understanding Older People's Experiences of Poverty and Material Deprivation. Department for Work and Pensions, Collingham.
- Downey, L., Van Willigen, M., 2005. Environmental stressors: the mental health impacts of living near industrial activity. *Journal of Health and Social Behavior* 46 (3), 289–305.
- Eager, S., Killaspy, H.C.J., Mezey, G., et al., 2023. Understanding the social inclusion needs of people living in mental health supported accommodation. *BMC Research Notes* 16 (1), 23289515.
- Eibich, P., Krekel, C., Demuth, I., et al., 2016. Associations between neighborhood characteristics, well-being and health vary over the life course. *Gerontology* 62 (3), 362–370.
- English National Concessionary Travel Scheme (ENCTS), 2017. Transport for NSW Disability Inclusion Action Plan 2018–2022. Available at: https://www.vgls.vic.gov.au/client/en_AU/search/asset/1297375/0 (Accessed 1 February 2024).
- Ensrud, K.E., Lui, L.Y., Paudel, M.L., et al., 2016. Effects of mobility and cognition on risk of mortality in women in late life: a prospective study. *The Journals of Gerontology Series A: Biological Sciences and Medical Sciences* 71 (6), 759–765.
- Ermagun, A., Tilahun, N., 2020. Equity of transit accessibility across Chicago. *Transportation Research Part D: Transport and Environment* 86, 102461.
- Estes, C.L., Binney, E.A., Culbertson, R.A., 1992. The gerontological imagination: social influences on the development of gerontology, 1945–present. *International Journal of Aging and Human Development* 35 (1), 49–65.
- Europe Commission, 2020. Poverty and Social Exclusion. Available at: <https://ec.europa.eu/social/main.jsp?catId=751&langId=en> (Accessed 1 February 2024).
- Feng, J., 2017. The influence of built environment on travel behavior of the elderly in urban China. *Transportation Research Part D: Transport and Environment* 52, 619–633.
- Fennel, G., Phillpson, C., Evers, H., 1993. *The Sociology of Old Age*. Open University Press, London.
- Fields, N.L., Cronley, C., Mattingly, S.P., et al., 2021. Transportation mobility and health among older adults: examining missed trips and latent demand. *Journal of Transport & Health* 21, 101069.
- Foster, L., Walker, A., 2014. Active and successful aging: a European policy perspective. *The Gerontologist* 55 (1), 83–90.
- Franke, T., Sims-Gould, J., Chaudhury, H., et al., 2019. It makes your life worthwhile. It gives you a purpose in living: mobility experiences among active older adults with low income. *Ageing and Society* 39 (8), 1639–1666.
- Frankfurt, H., 1987. Equality as a moral ideal. *Ethics* 98 (1), 21–43.
- Friman, M., Gärling, T., Ettema, D., et al., 2017. How does travel affect emotional well-being and life satisfaction? *Transportation Research Part A: Policy and Practice* 106 (1), 170–180.
- Gaglione, F., Cottrill, C., Gargiulo, C., 2021. Urban services, pedestrian networks and behaviors to measure elderly accessibility. *Transportation Research Part D: Transport and Environment* 90, 102687.
- Gandelman, N., Serebrisky, T., Suárez-Alemán, A., 2019. Household spending on transport in Latin America and the Caribbean: a dimension of transport affordability in the region. *Journal of Transport Geography* 79, 102482.
- Gao, Y., Rasoul, S., Timmermans, H., et al., 2018. Trip stage satisfaction of public transport users: a reference-based model incorporating trip attributes, perceived service quality, psychological disposition and difference tolerance. *Transportation Research Part A: Policy and Practice* 118, 759–775.
- Garnier, C., Trépanier, M., Morency, C., 2022. Criteria to prioritize opportunities to shift paratransit trips to regular transit network-Montreal case study. *Journal of Transport & Health* 24, 101338.
- Gawlak, A., Matuszewska, M., Ptak, A., 2021. Inclusiveness of urban space and tools for the assessment of the quality of urban life—a critical approach. *International Journal of Environmental Research and Public Health* 18 (9), 4519.
- Gayman, M.D., Turner, R.J., Cui, M., 2008. Physical limitations and depressive symptoms: exploring the nature of the association. *The Journal of Gerontology Series B: Psychological Sciences and Social Sciences* 63 (4), S219–S228.
- Graham, H., Bell, S., Flemming, K., et al., 2018. The experiences of everyday travel for older people in rural areas: a systematic review of UK qualitative studies. *Journal of Transport & Health* 11, 141–152.
- Graham, H., De Bell, S., Flemming, K., et al., 2020. Older people's experiences of everyday travel in the urban environment: a thematic synthesis of qualitative studies in the United Kingdom. *Ageing and Society* 40 (4), 842–868.
- Guagliardo, M.F., 2004. Spatial accessibility of primary care: concepts, methods and challenges. *International Journal of Health Geographics* 3 (1), 3.

- Guida, C., Carpentieri, G., Masoumi, H., 2022. Measuring spatial accessibility to urban services for older adults: an application to healthcare facilities in Milan. *European Transport Research Review* 14 (1), 23.
- Hagan, R.J., 2019. Getting out of the house: the use of community transport as a third place for rural-dwelling older adults. *Ageing and Society* 40 (11), 2519–2539.
- Hansen, W.G., 1959. How accessibility shapes land use. *Journal of the American Planning Association* 25 (2), 73–76.
- Harris, K.C., Semmler, B.J.F., Anderson, S., et al., 2024. Innovative solutions to support individuals with disabilities accessing public transportation: a case study. *Assistive Technology*, <https://doi.org/10.1080/10400435.2024.2305977>.
- He, S., Cheung, Y.H., Tao, S., 2018. Travel mobility and social participation among older people in a transit metropolis: a socio-spatial-temporal perspective. *Transportation Research Part A: Policy and Practice* 118, 608–626.
- He, Z., Ma, S., Tang, X., 2009. Empirical study on the influence of learning ability to individual travel behavior. *Journal of Transportation Systems Engineering and Information Technology* 9 (2), 75–80.
- He, S., Thgersen, J., Cheung, Y.H.Y., et al., 2020. Ageing in a transit-oriented city: satisfaction with transport, social inclusion and wellbeing. *Transport Policy* 97, 85–94.
- Heaven, B., Brown, L.J.E., White, M., et al., 2013. Supporting well-being in retirement through meaningful social roles: systematic review of intervention studies. *Milbank Quarterly* 91 (2), 222–287.
- Hemingway, A., Jack, E., Eleanor, Jack, 2013. Reducing social isolation and promoting well being in older people. *Quality in ageing: policy. Practice and Research* 14 (1), 25.
- Hernandez, D.O., Titheridge, H., 2016. Mobilities of the periphery: informality, access and social exclusion in the urban fringe in Colombia. *Journal of Transport Geography* 55 (3), 152–164.
- Hidayati, I.N., Tan, W., Yamu, C., 2021. Conceptualizing mobility inequality: mobility and accessibility for the marginalized. *Journal of Planning Literature* 36 (4), 492–507.
- Holmes, K., Clifford, S., Gregory, K., et al., 2007. ECLIPSE: Good Practice Review—Final Version Year Two. European Commission, Strasbourg.
- Holt-Lunstad, J., Smith, T.B., Baker, M.W., et al., 2015. Loneliness and social isolation as risk factors for mortality: a meta-analytic review. *Perspectives on Psychological Science* 10 (2), 227–237.
- Hou, Y., Moogoor, A., Dieterich, A.V., et al., 2021. Exploring built environment correlates of older adults' walking travel from lifelogging images. *Transportation Research Part D: Transport and Environment* 96, 102850.
- Hu, R., Dong, S., Zhao, Y., et al., 2013. Assessing potential spatial accessibility of health services in rural China: a case study of Donghai County. *International Journal for Equity in Health* 12 (1), 35.
- Hu, S., Song, W., Li, C., et al., 2020. A multi-mode Gaussian-based two-step floating catchment area method for measuring accessibility of urban parks. *Cities* 105, 102815.
- Ingvardson, J.B., Nielsen, O.A., 2019. The relationship between norms, satisfaction and public transport use: a comparison across six European cities using structural equation modelling. *Transportation Research Part A: Policy and Practice* 126, 37–57.
- International Council on Active Aging (ICAA), 2018. Mission and Vision. Available at: https://www.icaa.cc/about_us/misionvision.htm (Accessed 1 February 2024).
- Ivan, L., Dikken, J., Van Hoof, J., 2024. Unveiling the experienced age-friendliness of older people in Bucharest: a comprehensive study using the validated Romanian age-friendly cities and communities questionnaire and cluster analysis. *Habitat International* 143 (4), 102973.
- Jackson, S.E., Firth, J.A., Firth, J., et al., 2019. Social isolation and physical activity mediate associations between free bus travel and wellbeing among older adults in England. *Journal of Transport & Health* 13, 274–284.
- Jittrapirom, P., Caiati, V., Feneri, A.M., et al., 2017. Mobility as a service: a critical review of definitions. *Assessments of Schemes, and Key Challenges Urban Planning* 2 (2), 13–25.
- Johnson, R., Shaw, J., Berding, J., et al., 2017. European national government approaches to older people's transport system needs. *Transport Policy* 59, 17–27.
- Jones, A., Goodman, A., Roberts, H., et al., 2013. Entitlement to concessionary public transport and wellbeing: a qualitative study of young people and older citizens in London, UK. *Social Science & Medicine* 91, 202–209.
- Kamruzzaman, M., Yigitcanlar, T., Yang, J., et al., 2016. Measures of transport-related social exclusion: a critical review of the literature. *Sustainability* 8 (7), 696.
- Kandt, J., Leak, A., 2019. Examining inclusive mobility through smart card data: what shall we make of senior citizens' declining bus patronage in the west midlands? *Journal of Transport Geography* 79, 102474.
- Kaplan, S., Popoks, D., Prato, C.G., et al., 2014. Using connectivity for measuring equity in transit provision. *Journal of Transport Geography* 37, 82–92.
- Kębiowski, W., 2017. More than Just Riding without a Ticket? Exploring the Geography of Fare-free Public Transport (Cosmopolis Working Paper). Vrije Universiteit Brussel, Cosmopolis.
- Kelobonye, K., McCarney, G., Xia, J., et al., 2019. Relative accessibility analysis for key land uses: a spatial equity perspective. *Journal of Transport Geography* 75, 82–93.
- Kenyon, S., Lyons, G., Rafferty, J., 2002. Transport and social exclusion: investigating the possibility of promoting inclusion through virtual mobility. *Journal of Transport Geography* 10 (3), 207–219.
- Kim, J., Schmcker, J.D., Nakamura, T., et al., 2020. Integrated impacts of public transport travel and travel satisfaction on quality of life of older people. *Transportation Research Part A: Policy and Practice* 138 (1), 15–27.
- Kiss, M., 2022. Understanding Transport Poverty. European Parliamentary Research Service, European Union. Available at: [https://www.europarl.europa.eu/RegData/etudes/ATAG/2022/738181/EPRS_ATA\(2022\)738181_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/ATAG/2022/738181/EPRS_ATA(2022)738181_EN.pdf) (Accessed 1 February 2024).
- Kong, L., Woods, O., 2018. The ideological alignment of smart urbanism in Singapore: critical reflections on a political paradox. *Urban Studies* 55 (4), 679–701.
- Kovalchick, S., Camerer, C.F., Grether, D.M., et al., 2004. Aging and decision making: a comparison between neurologically healthy elderly and young individuals. *Journal of Economic Behavior and Organization* 58 (1), 79–94.
- Lanza, G., Pucci, P., Carboni, L., 2023. Measuring accessibility by proximity for an inclusive city. *Cities* 143, 104581.
- Lattman, K., Friman, M., Olsson, L.E., 2016a. Perceived accessibility of public transport as a potential indicator of social inclusion. *Social Inclusion* 4 (3), 36–45.
- Lattman, K., Olsson, L.E., Friman, M., 2016b. Development and test of the perceived accessibility scale (PAC) in public transport. *Journal of Transport Geography* 54, 257–263.
- Laverty, A.A., Webb, E., Vamos, E.P., et al., 2018. Associations of increases in public transport use with physical activity and adiposity in older adults. *International Journal of Behavioral Nutrition and Physical Activity* 15 (1), 31.
- Lee, S.M., Conway, T.L., Frank, L.D., et al., 2017. The relation of perceived and objective environment attributes to neighborhood satisfaction. *Environment and Behavior* 49 (2), 136–160.

- Lei, T., Church, R.L., 2010. Mapping transit-based access: integrating GIS, routes and schedules. *International Journal of Geographical Information Science* 24 (2), 283–304.
- Levasseur, M., G  n  reux, M., Bruneau, J.F., et al., 2015. Importance of proximity to resources, social support, transportation and neighborhood security for mobility and social participation in older adults: results from a scoping study. *BMC Public Health* 15 (1), 1–19.
- Li, J., Pei, H., Shi, X., et al., 2020a. A comparative study of quantitative measurement tools for active aging. *Chinese Journal of Evidence-based Medicine* 20 (3), 296–305.
- Li, C., Wang, J., 2024. Using an age-grouped Gaussian-based two-step floating catchment area method (AG2SFCA) to measure walking accessibility to urban parks: with an explicit focus on elderly. *Journal of Transport Geography* 114, 103772.
- Li, J., Xu, L., Yao, D., et al., 2019. Impacts of symbolic value and passenger satisfaction on bus use. *Transportation Research Part D: Transport and Environment* 72, 98–113.
- Li, S., Zhang, Y., Ruan, H., et al., 2020b. The role of transportation in older adults' use of and satisfaction with primary care in China. *Journal of Transport and Health* 18, 100898.
- Liang, H., Yan, Q., Yan, Y., et al., 2023. Using an improved 3SFCA method to assess inequities associated with multi-modal accessibility to green spaces based on mismatches between supply and demand in the metropolitan of Shanghai, China. *Sustainable Cities and Society* 91, 104456.
- Liu, Q., An, Z., Liu, Y., et al., 2021a. Smartphone-based services, perceived accessibility, and transport inequity during the COVID-19 pandemic: a cross-lagged panel study. *Transportation Research Part D: Transport and Environment* 97, 102941.
- Liu, C., Bardaka, E., Palakurthy, R., et al., 2020. Analysis of travel characteristics and access mode choice of elderly urban rail riders in Denver, Colorado. *Travel Behaviour and Society* 19, 194–206.
- Liu, Q., Liu, Y., Zhang, C., et al., 2021b. Elderly mobility during the COVID-19 pandemic: a qualitative exploration in Kunming, China. *Journal of Transport Geography* 96, 103176.
- Liu, J., Rozelle, S., Xu, Q., et al., 2019. Social engagement and elderly health in China: evidence from the China health and retirement longitudinal survey (CHARLS). *International Journal of Environmental Research and Public Health* 16 (2), 278.
- Liu, J., Zhong, S., Huang, Y., et al., 2021c. How does the preference heterogeneity affect the elderly's evaluation of bus accessibility? Evidence from Guangzhou, China. *Journal of Transport & Health* 22, 101094.
- Lucas, K., 2004. *Transport and Social Exclusion: a Survey of the Group of Seven Nations*. FIA Foundation, London.
- Lucas, K., 2012. Transport and social exclusion: where are we now? *Transport Policy* 20, 105–113.
- Lucas, K., 2019. A new evolution for transport-related social exclusion research? *Journal of Transport Geography* 81, 102529.
- Lucas, K., Jones, P., 2012. Social impacts and equity issues in transport: an introduction. *Journal of Transport Geography* 21, 1–3.
- Lucas, K., Markovich, J., 2011. International perspectives. In: Currie, G. (Ed.), *New Perspectives and Methods in Transport and Social Exclusion Research*. Emerald, Bradford, pp. 223–239.
- Lucas, K., Mattioli, G., Verlinghieri, E., et al., 2016. Transport poverty and its adverse social consequences. *Transport* 169 (6), 353–365.
- Luiu, C., Tight, M., 2021. Travel difficulties and barriers during later life: evidence from the national travel survey in England. *Journal of Transport Geography* 91, 102973.
- Luiu, C., Tight, M.R., Burrow, M., 2017. The unmet travel needs of the older population: a review of the literature. *Transport Reviews* 37 (4), 488–506.
- Luz, G., Portugal, L., 2021. Understanding transport-related social exclusion through the lens of capabilities approach. *Transport Reviews* 42 (4), 503–525.
- Ma, T., Ivers, R., Solar, J., et al., 2022. Experiences of older aboriginal people in navigating transport systems in an urban setting: an indigenous perspective on transport access, a social determinant of health. *International Journal of Environmental Research and Public Health (IJERPH)* 19 (21), 13778.
- Ma, L., Kent, J., Mulley, C., 2018. Transport disadvantage, social exclusion, and subjective well-being: the role of the neighborhood environment-evidence from Sydney, Australia. *Journal of Transport and Land Use* 11 (1), 31–47.
- Macintyre, S., Macdonald, L., Ellaway, A., 2008. Do poorer people have poorer access to local resources and facilities? The distribution of local resources by area deprivation in Glasgow, Scotland. *Social Science & Medicine* 67 (6), 900–914.
- Mackett, R.L., 2014a. Has the policy of concessionary bus travel for older people in Britain been successful? *Case Studies on Transport Policy* 2 (2), 81–88.
- Mackett, R.L., 2014b. The health implications of inequalities in travel. *Journal of Transport & Health* 1 (3), 202–209.
- Mackett, R.L., Thoreau, R., 2015. Transport, social exclusion and health. *Journal of Transport & Health* 2 (4), 610–617.
- Mackey, D.C., Lui, L.Y., Cawthon, P.M., et al., 2016. Life-space mobility and mortality in older women: prospective results from the study of osteoporotic fractures. *Journal of the American Geriatrics Society* 64 (11), 2226–2234.
- Mah, S., Mitra, R., 2017. The effects of a free bus program on older adults travel behavior: a case study of a Canadian suburban municipality. *Case Studies on Transport Policy* 5 (3), 460–466.
- Malekzadeh, A.R., Chung, E., 2020. A review of transit accessibility models: challenges in developing transit accessibility models. *International Journal of Sustainable Transportation* 14 (10), 733–748.
- Martens, K., 2012. Justice in transport as justice in accessibility: applying Walzer's "spheres of justice" to the transport sector. *Transportation* 39, 1035–1053.
- Martens, K., Golub, A., Robinson, G., 2012. A justice-theoretic approach to the distribution of transportation benefits: implications for transportation planning practice in the United States. *Transportation Research Part A: Policy and Practice* 46 (4), 684–695.
- Marx, A., Rihoux, B., Ragin, C., 2014. The origins, development, and application of qualitative comparative analysis: the first 25 years. *European Political Science Review* 6 (1), 115–142.
- Massey, D.S., Mullan, B.P., 1984. Processes of Hispanic and black spatial assimilation. *American Journal of Sociology* 89 (4), 836–873.
- Mavoa, S., Witten, K., McCreanor, T., et al., 2012. GIS based destination accessibility via public transit and walking in Auckland, New Zealand. *Journal of Transport Geography* 20 (1), 15–22.
- McDonagh, J., 2006. Transport policy instruments and transport-related social exclusion in rural Republic of Ireland. *Journal of Transport Geography* 14 (5), 355–366.
- Midlarsky, E., 1991. Helping as coping. In: Clark, M.S. (Ed.), *Prosocial Behavior*. Sage, Newbury Park, pp. 238–264.
- Mirzoev, T., Tull, K.I., Winn, N., et al., 2021. Systematic review of the role of social inclusion within sustainable urban developments. *International Journal of Sustainable Development and World Ecology* 29 (1), 3–17.
- Misra, A., Shirgaokar, M., Agrawal, A.W., et al., 2022. How older adults use ride-hailing booking technology in California. *Transportation Research Part A: Policy and Practice* 155, 11–30.

- Mitra, S., Yao, M., Ritchie, S.G., 2021. Gender differences in elderly mobility in the United States. *Transportation Research Part A: Policy and Practice* 154 (3), 203–226.
- Mokhtarian, P.L., 2005. Travel as a desired end, not just a means. *Transportation Research Part A: Policy and Practice* 39 (2–3), 93–96.
- Morioka, W., Kwan, M.P., Hino, K., et al., 2023. How accessibility to neighborhood grocery stores is related to older people's walking behavior: a study of Yokohama, Japan. *Journal of Transport and Health* 32, 101668.
- Morris, A., Barnes, J., Fildes, B., 2017. Challenges of using the bus as an older person. *WIT Transactions on the Built Environment* 176, 539–544.
- Mouratidis, K., 2021. Urban planning and quality of life: a review of pathways linking the built environment to subjective well-being. *Cities* 115, 103229.
- Musich, S., Wang, S., Ruiz, J., et al., 2018. The impact of mobility limitations on health outcomes among older adults. *Geriatric Nursing* 39 (2), 162–169.
- Musselwhite, C., 2015. Further examinations of mobility in later life and improving health and wellbeing. *Journal of Transport & Health* 2 (2), 99–100.
- Musselwhite, C., Haddad, H., 2010. Mobility, accessibility and quality of later life. *Quality in Ageing and Older Adults* 11 (1), 25–37.
- Nápoles, V.M.P., Páez, D.G., Penelas, J.L., et al., 2020. Smart bus stops as interconnected public spaces for increasing social inclusiveness and quality of life of elder users. *Smart Cities* 3 (2), 430–443.
- Neves, B.D., Unsworth, C.A., Browning, C., 2023. “Being treated like an actual person”: attitudinal accessibility on the bus. *Mobilities* 18 (3), 425–444.
- Nikitas, A., Avineri, E., Parkhurst, G., 2018. Understanding the public acceptability of road pricing and the roles of older age, social norms, pro-social values and trust for urban policy-making: the case of Bristol. *Cities* 79, 78–91.
- Nordbakke, S., Schwanen, T., 2014. Well-being and mobility: a theoretical framework and literature review focusing on older people. *Mobilities* 9 (1), 104–129.
- Northern Ireland Statistics and Research Agency, 2005. The Northern Ireland Multiple Deprivation Measure. Northern Ireland Statistics and Research Agency (NISRA), Belfast.
- Nuworsoo, C., Golub, A., Deakin, E., 2009. Analyzing equity impacts of transit fare changes: case study of Alameda-Contra Costa Transit, California. *Evaluation and Program Planning* 32 (4), 360–368.
- Olsson, L.E., Friman, M., Lättman, K., et al., 2020. Travel and life satisfaction: from Gen Z to the silent generation. *Journal of Transport & Health* 18, 100894.
- O'Sullivan, D., Morrison, A., Shearer, J., 2000. Using desktop GIS for the investigation of accessibility by public transport: an isochrone approach. *International Journal of Geographical Information Science* 14 (1), 85–104.
- Özkazanç, S., Sönmez, F.N.O., 2017. Spatial analysis of social exclusion from a transportation perspective: a case study of Ankara metropolitan area. *Cities* 67, 74–84.
- Pantelaki, E., Maggi, E., Crotti, D., 2021. Mobility impact and well-being in later life: a multidisciplinary systematic review. *Research in Transportation Economics* 86, 100975.
- Paraschiv, D.M., Ioana, M.D., Țițan, E., et al., 2021. Development of an aggregated social inclusion indicator. Disparities in the European Union on inclusion/exclusion with social inclusion index. *Technological and Economic Development of Economy* 27 (6), 1–24.
- Patil, D.S., Bailey, A., Yadav, U.N., et al., 2022. Contextual factors influencing the urban mobility infrastructure interventions and policies for older adults in low- and middle-income countries: a realist review. *BMC Public Health* 22 (1), 1489.
- Peek, S.T.M., Wouters, E.J.M., Van Hoof, J., et al., 2014. Factors influencing acceptance of technology for aging in place: a systematic review. *International Journal of Medical Informatics* 83 (4), 235–248.
- Pérez-Cuevas, R., Doubova, S.V., Bazaldúa-Merino, L.A.B., et al., 2015. A social health services model to promote active aging in Mexico: design and evaluation of a pilot programme. *Ageing and Society* 35 (7), 1457–1480.
- Pfaller, L., Schweda, M., 2019. Excluded from the good life? An ethical approach to conceptions of active aging. *Social Inclusion* 7 (3), 44–53.
- Plushteva, A., Boussauw, K., 2020. Does night-time public transport contribute to inclusive night mobility? Exploring Sofia's night bus network from a gender perspective. *Transport Policy* 87, 41–50.
- Popay, J., Escorel, S., Hernández, M., et al., 2008. Understanding and Tackling Social Exclusion: Final Report to the WHO Commission on Social Determinants of Health from the Social Exclusion Knowledge Network. WHO, Geneva.
- Prescott, M., Labbé, D., Miller, W.C., et al., 2020. Factors that affect the ability of people with disabilities to walk or wheel to destinations in their community: a scoping review. *Transport Reviews* 40 (5), 646–669.
- Prince, M.J., Wu, F., Guo, Y., et al., 2015. The burden of disease in older people and implications for health policy and practice. *The Lancet* 385 (9967), 549–562.
- Radke, J., Mu, L., 2000. Spatial decompositions, modeling and mapping service regions to predict access to social programs. *Geographic Information Sciences* 6 (2), 105–112.
- Rajtar-Zembaty, A., Rajtar-Zembaty, J., Saikowski, A., et al., 2019. Global cognitive functioning and physical mobility in older adults with and without mild cognitive impairment: evidence and implications. *Folia Medica Cracoviensia* 59 (1), 75–88.
- Ramachandran, M., D'souza, S.A., 2015. A cross-sectional survey on older adults' community mobility in an Indian metropolis. *Journal of Cross-Cultural Gerontology* 31, 19–33.
- Rambaldini-Gooding, D., Molloy, L., Parrish, A.M., et al., 2021. Exploring the impact of public transport including free and subsidised on the physical, mental and social well-being of older adults: a literature review. *Transplant Review* 41 (5), 600–616.
- Rambaldini-Gooding, D., Molloy, L., Parrish, A.M., et al., 2023. Free public transport and older people: an ethnographic study of an Australian bus service. *Australasian Journal on Ageing* 42 (2), 409–416.
- Ramia, I., Voicu, M., 2022. Life satisfaction and happiness among older Europeans: the role of active aging. *Social Indicators Research* 160, 667–687.
- Ramjerdi, F., 2006. Equity measures and their performance in transportation. *Transportation Research Board* 1983, 67–74.
- Ramsetty, A., Adams, C., 2020. Impact of the digital divide in the age of COVID-19. *Journal of the American Medical Informatics Association* 27 (7), 1147–1148.
- Randal, E., Shaw, C., Woodward, A., et al., 2020. Fairness in transport policy: a new approach to applying distributive justice theories. *Sustainability* 12 (23), 10102.
- Rastogi, R., Rao, K.V.K., 2003. Defining transit accessibility with environmental inputs. *Transportation Research Part D: Transport and Environment* 8 (5), 383–396.
- Rawls, J., 1971. *A Theory of Justice*. Harvard University Press, Cambridge.
- Raymond, M., Grenier, A., Hanley, J., 2014. Community participation of older adults with disabilities. *Journal of Community & Applied Social Psychology* 24 (1), 50–62.
- Reinhard, E., Carrino, L., Courtin, E., et al., 2019. Public transportation use and cognitive function in older age: a quasi-experimental evaluation of the free bus pass policy in

- the United Kingdom. *American Journal of Epidemiology* 188 (10), 1774–1783.
- Reinhard, E., Courtin, E., Van Lenthe, F.J., et al., 2018. Public transport policy, social engagement and mental health in older age: a quasi-experimental evaluation of free bus passes in England. *Journal of Epidemiology and Community Health* 72 (5), 361–368.
- Ricciardi, A.M., Xia, J., Currie, G., 2015. Exploring public transport equity between separate disadvantaged cohorts: a case study in Perth, Australia. *Journal of Transport Geography* 43, 111–122.
- Rojas, C., Paez, A., Barbosa, O., et al., 2016. Accessibility to urban green spaces in Chilean cities using adaptive thresholds. *Journal of Transport Geography* 57, 227–240.
- Rosso, A.L., Taylor, J.A., Tabb, L.P., et al., 2013. Mobility, disability, and social engagement in older adults. *Journal of Aging and Health* 25 (4), 617–637.
- Ryan, J., Wretstrand, A., Schmidt, S.M., 2019. Disparities in mobility among older people: findings from a capability-based travel survey. *Transport Policy* 79, 177–192.
- Rye, T., 2008. *The Implementation and Effectiveness of Transport Demand Management Measures: an International Perspective*, first ed. Routledge, London.
- Rye, T., Carreno, M., 2008. Concessionary fares and bus operator reimbursement in Scotland and Wales: no better or no worse off? *Transport Policy* 15 (4), 242–250.
- Rye, T., Mykura, W., 2009. Concessionary bus fares for older people in Scotland are they achieving their objectives. *Journal of Transport Geography* 17, 451–456.
- Santos, L.J., Oliveira, A.V.M., Aldrighi, D.M., 2021. Testing the differentiated impact of the COVID-19 pandemic on air travel demand considering social inclusion. *Journal of Air Transport Management* 94, 102082.
- Saz-Gil, I., Gil-Lacruz, A.I., Gil-Lacruz, M., 2019. Active aging and happiness: European senior citizen volunteers. *Sociologia-Slovak Sociological Review* 51 (3), 290–312.
- Scottish Executive, 2006. *Scottish Index of Multiple Deprivation 2006: General Report*. Scottish Executive, Edinburgh.
- Selwyn, N., 2004. The information aged: a qualitative study of older adults' use of information and communications technology. *Journal of Aging Studies* 18 (4), 369–384.
- Sen, A.K., 1992. *Inequality Reexamined*. Clarendon, Oxford.
- Shankar, A., McMunn, A., Demakakos, P., et al., 2017. Social isolation and loneliness: prospective associations with functional status in older adults. *Health Psychology* 36 (2), 179–187.
- Sharaby, N., Shiftan, Y., 2012. The impact of fare integration on travel behavior and transit ridership. *Transport Policy* 21, 63–67.
- Sharma, I., Mishra, S., Golias, M.M., et al., 2020. Equity of transit connectivity in Tennessee cities. *Journal of Transport Geography* 86, 102750.
- Shirgaokar, M., 2020. Expanding seniors' mobility through phone apps: potential responses from the private and public sectors. *Journal of Planning Education and Research* 40 (4), 405–415.
- Shookner, M., 2002. *An Inclusion Lens: Workbook for Looking at Social and Economic Exclusion and Inclusion*. Population and Public Health Branch, Atlantic Region, Health Canada, Halifax.
- Shrestha, B.P., McDonald, M., Millonig, A., et al., 2017. Review of public transport needs of older people in European context. *Journal of Population Ageing* 10 (4), 343–361.
- Singleton, P.A., 2019. Validating the satisfaction with travel scale as a measure of hedonic subjective well-being for commuting in a U.S. city. *Transportation Research Part F: Traffic Psychology and Behaviour* 60 (6), 399–414.
- Siren, A., Hjorthol, R., Levin, L., 2015. Different types of out-of-home activities and wellbeing among urban residing old persons with mobility impediments. *Journal of Transport & Health* 2 (1), 14–21.
- Sochor, J., 2014. The reassuring effects of ICT in public transportation: the perspectives of older adults. In: 93rd Annual Meeting of the Transportation Research Board, Washington DC, 2014.
- Sourbati, M., Behrendt, F., 2020. Smart mobility, age and data justice. *New Media & Society* 23 (6), 1398–1414.
- Stanley, J., 2004. *Improving Public Transport to Meet Community Needs: a Warrnambool Case-Study*. Bus Association Victoria and Warrnambool Bus Lines, Melbourne.
- Stanley, J., Hensher, D.A., Stanley, J.R., et al., 2011a. Mobility, social exclusion and well-being: exploring the links. *Transportation Research Part A: Policy and Practice* 45 (8), 789–801.
- Stanley, J., Hensher, D.A., Stanley, J.R., et al., 2011b. Social exclusion and the value of mobility. *Journal of Transport Economics and Policy* 45 (2), 197–222.
- Stanley, J., Vella-Brodick, D., 2009. The usefulness of social exclusion to inform social policy in transport. *Transport Policy* 16 (3), 90–96.
- Stjernborg, V., Svensson, G.L., 2024. Rural mobility in later life: counteracting accessibility poverty with digital service solutions. *Transportation Research Part D: Transport and Environment* 126, 104030.
- Su, F., Bell, M.G.H., 2009. Transport for older people: characteristics and solutions. *Research in Transportation Economics* 25 (1), 46–55.
- Sundling, C., Emardson, R., Pendrill, L.R., et al., 2015. Two models of accessibility to railway traveling for vulnerable elderly persons. *Measurement* 72, 96–101.
- Swift, A., Cheng, L., Loo, B.P.Y., et al., 2021. Step-free railway station access in the UK: the value of inclusive design. *European Transport Research Review* 13, 1–12.
- Sze, N.N., Christensen, K.M., 2017. Access to urban transportation system for individuals with disabilities. *IATSS Research* 41 (2), 66–73.
- Szeto, W.Y., Yang, L., Wong, R.C.P., et al., 2017. Spatio-temporal travel characteristics of the elderly in an aging society. *Travel Behaviour and Society* 9, 10–20.
- Thomopoulos, N., 2013. Transport and ethics: ethics and the evaluation of transport policies and projects. *Transport Reviews* 33 (4), 497–498.
- Tiznado-Aitken, I., Muñoz, J.C., Hurtubia, R., 2021. Public transport accessibility accounting for level of service and competition for urban opportunities: an equity analysis for education in Santiago de Chile. *Journal of Transport Geography* 90, 102919.
- Tucker, R., Kelly, D., Frawley, P., et al., 2022. Making an Australian regional city accessible and inclusive: a transdisciplinary research approach to community participation in urban. *Urban Policy and Research* 41 (2), 1–17.
- Turbay, A.L.B., Firmino, R.J., 2023. Mobility as a Service as an instrument to mitigate transport-related social exclusion. *Revista Tecnologia e Sociedade* 19 (56), 72.
- UK Parliament, 2020. *Research Briefing: Concessionary Bus Travel*. Available at: <https://commonslibrary.parliament.uk/research-briefings/sn01499/#:~:text=The%20English%20National%20Concessionary%20Travel,weekends%20and%20on%20Bank%20Holidays> (Accessed 1 February 2024).
- United Nations, 2017. *World Population Aging 2017*. United Nations Publication, New York.
- Van Cauwenberg, J., De Bourdeaudhuij, I., Clarys, P., et al., 2018. Do electric bicycles contribute to active ageing? *Journal of Transport & Health* 9 (S), S3–S4.
- Van Lange, P.A.M., Otten, W., De Bruin, E.M.N., et al., 1997. Development of prosocial, individualistic and competitive orientations: theory and preliminary evidence. *Journal of Personality and Social Psychology* 73 (4), 733–746.

- Van Oorschot, W., Arts, W., Halman, L., 2005. Welfare state effects on social capital and informal solidarity in the European Union: evidence from the 1999/2000 European values study. *Policy and Politics* 33 (1), 33–54.
- Van Wee, B., 2016. Accessible accessibility research challenges. *Journal of Transport Geography* 51, 9–16.
- Vanolo, A., 2014. Smart mentality: the smart city as disciplinary strategy. *Urban Studies* 51 (5), 883–898.
- Varghese, V., Jana, A., 2019. Interrelationships between ICT, social disadvantage, and activity participation behaviour: a case of Mumbai, India. *Transportation Research Part A: Policy and Practice* 125, 248–267.
- Vecchio, G., Martens, K., 2021. Accessibility and the capabilities approach: a review of the literature and proposal for conceptual advancements. *Transport Reviews* 41 (6), 833–854.
- Vella-Brodrick, D.A., Stanley, J., 2013. The significance of transport mobility in predicting well-being. *Transport Policy* 29, 236–242.
- Verhorst, T., Fu, X., Van Lierop, D., 2023. Definitions matter: investigating indicators for transport poverty using different measurement tools. *European Transport Research Review* 15, 1–17.
- Villena-Sanchez, J., Boschmann, E.E., Ávila-Forcada, S., 2022. Daily travel behaviors and transport mode choice of older adults in Mexico City. *Journal of Transport Geography* 104 (2), 103445.
- Vine, D., Buys, L., Aird, R., 2012. Experiences of neighbourhood walk ability among older Australians living in high density inner-city areas. *Planning Theory and Practice* 13 (3), 421–444.
- Voss, C., Sims-Gould, J., Ashe, M.C., et al., 2016. Public transit use and physical activity in community-dwelling older adults: combining GPS and accelerometry to assess transportation-related physical activity. *Journal of Transport & Health* 3 (2), 191–199.
- Walzer, M., 1983. *Spheres of Justice: a Defense of Pluralism and Equality*. Basic Books, New York.
- Wang, C., 2009. Dialectics in multitude an exploration into/ beyond Henri Lefebvre's conceptual triad of production of space. *Journal of Geographical Science* 55, 1–24.
- Wang, C., Chen, N., 2015. A GIS-based spatial statistical approach to modeling job accessibility by transportation mode: case study of Columbus, Ohio. *Journal of Transport Geography* 45, 1–11.
- Wang, Y., Wong, Y.D., Goh, K., 2019. Perceived importance of inclusive street dimensions: a public questionnaire survey from a vision(ing) perspective. *Transportation* 48 (5), 699–721.
- Webb, E., Laverty, A., Mindell, J., et al., 2016. Free bus travel and physical activity, gait speed, and adiposity in the English longitudinal study of aging. *American Journal of Public Health* 106 (1), 136–142.
- Webb, E., Netuveli, G., Millett, C., 2012. Free bus passes, use of public transport and obesity among older people in England. *Journal of Epidemiology and Community Health* 66 (2), 176–180.
- Welch, T.F., Mishra, S., 2013. A measure of equity for public transit connectivity. *Journal of Transport Geography* 33, 29–41.
- Welsh Assembly Government, 2008. *Welsh Index of Multiple Deprivation 2008 (WIMD)–Summary Report*. Welsh Assembly Government, Caernarfon.
- Whelan, M., Langford, J., Oxley, J., et al., 2006. The elderly and mobility: a review of the literature. *Accident Research Centre* 255, 88.
- Wiener, J.M., Büchner, S.J., Hölscher, C., 2009. Taxonomy of human wayfinding tasks: a knowledge-based approach. *Spatial Cognition Computation* 9 (2), 152–165.
- Windle, G., Burholt, V., 2003. Older people in Wales, their transport and mobility: a literature review. *Quality in Ageing and Older Adults* 4 (2), 28–35.
- World Health Organization, 2002. *Active Ageing: a Policy Framework*. World Health Organization, Geneva.
- Wu, W., Zheng, T., 2023. Establishing a “dynamic two-step floating catchment area method” to assess the accessibility of urban green space in Shenyang based on dynamic population data and multiple modes of transportation. *Urban Forestry and Urban Greening* 82, 127893.
- Yang, Y., Langellier, B.A., Stankov, I., et al., 2020. Public transit and depression among older adults: using agent-based models to examine plausible impacts of a free bus policy. *Journal of Epidemiology and Community Health* 74 (11), 875–881.
- Yigitcanlar, T., Dur, F., Dizdaroglu, D., 2015. Towards prosperous sustainable cities: a multiscale urban sustainability assessment approach. *Habitat International* 45, 36–46.
- Yigitcanlar, T., Mohamed, A., Kamruzzaman, M., et al., 2019. Understanding transport-related social exclusion: a multidimensional approach. *Urban Policy and Research* 37 (1), 97–110.
- Yuan, Y., Yang, M., Wu, J., et al., 2019. Assessing bus transit service from the perspective of elderly passengers in Harbin, China. *International Journal of Sustainable Transportation* 13 (10), 761–776.
- Zeitler, E., Buys, L., 2015. Mobility and out-of-home activities of older people living in suburban environments: “because I’m a driver, I don’t have a problem”. *Ageing and Society* 35 (4), 785–808.
- Zhang, J., Cheng, Y., Zhao, B., 2021. Assessing the inequities in access to peri-urban parks at the regional level: a case study in China’s largest urban agglomeration. *Urban Forestry and Urban Greening* 65, 127334.
- Zhang, Y., Yao, E., Zhang, R., et al., 2019. Analysis of elderly people’s travel behaviours during the morning peak hours in the context of the free bus programme in Beijing, China. *Journal of Transport Geography* 76, 191–199.
- Zhang, S., Yu, P., Chen, Y., et al., 2022. Accessibility of park green space in Wuhan, China: implications for spatial equity in the post-COVID-19 era. *International Journal of Environmental Research and Public Health* 19 (9), 5440.
- Zhang, N., Zhang, J., Yang, Q., et al., 2024. The impact of transport inclusion on active aging: a perceived value analysis. *Transportation Research Part D: Transport and Environment* 127, 104029.
- Zhang, M., Zhao, P., 2021. Literature review on urban transport equity in transitional China: from empirical studies to universal knowledge. *Journal of Transport Geography* 96, 103177.
- Zhang, M., Zhao, P., Qiao, S., 2020. Smartness-induced transport inequality: privacy concern, lacking knowledge of smartphone use and unequal access to transport information. *Transport Policy* 99, 175–185.
- Zhao, Y., Yamamoto, T., Kanamori, R., 2020. Study of older male drivers’ driving stress compared with that of young male drivers. *Journal of Traffic and Transport Engineering (English Edition)* 7 (4), 467–481.
- Zheng, Z., Shen, W., Li, Y., et al., 2020. Spatial equity of park green space using KD2SFCA and web map API: a case study of Zhengzhou, China. *Applied Geography* 123, 102310.
- Zhou, J., Zhang, M., Zhu, P., 2019. The equity and spatial implications of transit fare. *Transportation Research Part A: Policy and Practice* 121, 309–324.
- Ziegler, F., Schwanen, T., 2011. “I like to go out to be energised by different people”: an exploratory analysis of mobility and well being in later life. *Ageing and Society* 31 (5), 758–781.



Na Zhang is an associate professor and a PhD student at Chang'an University. Her research interests include travel behavior, public transport and transport culture.



Dr. Qi Yang is a professor at School of Economics and Management, Chang'an University. His research interests include transportation planning and management, transportation services, transportation economy, logistics and supply chain management.